

Daily Briefing - November 10, 2026

Your Daily Tech & Programming Digest

Tuesday, November 10, 2026

1000	116906	1146	38
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Automatic computational reproducibility checks with Metacheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-10-30 📖 19 min 📝 3854 words

TWENTY PERCENT STATISTICIAN

> **Summary:** <p>When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

> **Summary:** The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Dynamic muscle proprioception impairment in adolescent idiopathic scoliosis revealed by tendon vibration

📖 1 min 📝 29 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Daniela M Pinzon-Corredor, Christophe Boulay, Alexis Ulian, Guillaume Authier, Nicole Levy, Fabienne Levy, Jean-Luc Jouve, Corentin Petitpas, Sébastien Pesenti, Anne Kavounoudias

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007165?dgcid=rss_sd_all

Altered stability of dynamic functional architecture in the co-occurrence of autism spectrum disorder and attention–deficit/hyperactivity disorder

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Shuting Zhang, Chao Han, Yuning Si, Jinsha Tan, Xin Pan, Kaixin Li, Mengzhu Wang, Xipeng Yue, Xuan Yu, Wei Wei, Yan Bai, Meiyun Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007049?dgcid=rss_sd_all

Brain-heart decoupling in somatic depression: Desynchronization between salience network neuropsychological connectivity and heart rate variability

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yi Xia, Junling Sheng, Xiaoqin Wang, Ciqing Bao, Yujue Wang, Xumiao Wang, Zhijian Yao, Qing Lu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006464?dgcid=rss_sd_all

Do autistic traits modulate the neural processing of logical rule violations in visual and auditory modalities?

📖 1 min 📝 28 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Junyu Wu, Rong Huang, Jingyang Yu, Jiayi Shi, Tianze Wang, Yuting Zhang, Yutong Chen, Jie Li, Shubo Shui, Jing Meng

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006427?dgcid=rss_sd_all

Training interbrain synchrony through dyadic neurofeedback modulates empathy-for-pain

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Mario Francis, Simone Shamay-Tsoory

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006415?dgcid=rss_sd_all

Short-video experience and brain-state dynamics during the processing of rapidly changing video information in youth

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Bijie Tie, Li Geng, Pengcheng Wang, Jason M. Nagata, Jinbo He, Yueyang Xiao, Xueyang Wang, Jiang Qiu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600649X?dgcid=rss_sd_all

Widespread oscillatory suppression in autism persists in the absence of psychotropic medication.

👤 Amorim, A. K. J., Alves, G. M., Pontes, G. C. L., Dias, L., Hondermann, B. M. F., Silva, M. C. C., Costa, L., Ceriatte, C., Pereira, A., Brasil, L. M. B. F., Souza, I. C. N., Machado, C. L. M. V., Lima, B.

📅 2026-11-10 📖 1 min 📝 240 words BIORXIV NEUROSCIENCE

> **Summary:** Purpose: Investigate the neural oscillatory activity associated with autism spectrum disorder (ASD) using electroencephalographic (EEG) recordings in children and adolescents. Compare the corresponding oscillatory power between ASD patients taking psychotropic medication with a drug-naive group. Met...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.07.764864v1?rss=1

Accessing Mesoscale Functional Organization from Standard Functional MRI Leveraging AI-Based Super-Resolution

Nasr, S., Li, H. B., Zemlyanker, D., Huber, L., Rosen, M. S., Iglesias, J. E.

2026-11-10 1 min

185 words

BIORXIV NEUROSCIENCE

Summary: Access to the fine-scale functional organization of the human brain is essential for understanding cognition and neurological disorders yet remains beyond the reach of conventional functional MRI (fMRI) because of its limited spatial resolution and contrast-to-noise ratio. To overcome this limitatio...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.07.765069v1?rss=1

Rab10 and Arhgef7 interact to promote axon extension by linking vesicle supply and exocytosis

Olocco, F., Bekker, E., Ravindran, P., Pietralla, T., Puschel, A. W.

2026-11-10 1 min

175 words

BIORXIV NEUROSCIENCE

Summary: The formation of functional neural circuits begins with the extension of a single axon by immature neurons. Axon elongation requires an expansion of the plasma membrane that depends on the polarization of vesicle trafficking and exocytosis. It is not completely understood how the increased transport...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.764424v1?rss=1

Maternal separation remodels Purkinje cell circuits and impairs cerebellar synaptic plasticity

👤 Lippiello, P., Hoxha, E., Cristiano, C., Pascarella, S., Volpicelli, F., Speranza, L., Cuomo, M., Ferraro, M. G., Stanley, A., Rominto, A. M., Filiz, K. D., Liguori, F., Tropea, D., Di Angelantonio, S., Chiariotti, L., Sulzer, D., Miniaci, M. C.

📅 2026-11-10 📖 1 min 📝 243 words BIORXIV NEUROSCIENCE

Summary: Introduction: Early-life adversity has been associated with cerebellar abnormalities, but the cellular and circuit-level changes involved and their persistence across development remain poorly understood. We examined whether maternal separation (MS) disrupts cerebellar circuit maturation and whether...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.08.765140v1?rss=1

Multimodal Reorganization of Gut--Brain Physiology During the Cephalic Phase of Digestion

👤 Kehnemouyi, Y. M., Coleman, T. P.

📅 2026-11-10 📖 1 min 📝 205 words

BIORXIV NEUROSCIENCE

Summary: The vagus nerve plays a central role in gut--brain communication, yet the physiological dynamics associated with cholinergic vagal signaling remain difficult to characterize non-invasively in humans. The cephalic phase response, which prepares the gastrointestinal system for food ingestion through a...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.07.765106v1?rss=1

Subcellular Spatial Profiling Unveils Links Between Neuritic Plaques with Selective Tau Vulnerability in Alzheimer's Disease

👤 Li, H., Gross, E. R., Devine, A., Gundavelli, A., Prell, S. A., Woodbury, M. E., Gen, L., Doering, A., Held, A., Hyman, B. T., Serrano-Pozo, A., Zwang, T. J., Bennett, R. E., Das, S.

📅 2026-11-10 📖 1 min 📝 147 words BIORXIV NEUROSCIENCE

Summary: Alzheimer's disease (AD) features widespread extracellular amyloid-{beta} plaques but selective intraneuronal tau accumulation in cortical layers II/III and V, while layer IV remains relatively resistant. Here, we combined subcellular spatial transcriptomics with co-registered A{beta} and pTau immun...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.05.764692v1?rss=1

Tendon-specific PIEZO1 activation enhances neural drive to muscle during ballistic movement

Nakamura, K., Kosugi, A., Nakamichi, R., Kurimoto, R., Matsushima, T., Chiba, T., Ko, M., Morita, T., Yoshii, T., Seki, K., Asahara, H.

2026-11-10 1 min 250 words BIORXIV NEUROSCIENCE

Summary: Tendons are generally regarded as passive structures that transmit muscle force to bone, yet they contain the Golgi tendon organ (GTO), whose Ib afferents act on motor neurons either inhibitory or excitatory depending on movement context. Whether a change in tendon mechanical properties alters the n...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.07.765061v1?rss=1

Physical activity, fitness, sedentary behaviour, and brain health: a systematic review of Mendelian Randomisation studies

2026-11-10 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s44437-026-00013-2

Temporal dynamics of premenstrual dysphoric disorder revealed through predictive symptom networks

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78590-x

Social status and status loss do not influence alcohol-related behaviors in rats

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-80056-5

Some people are resistant to dementia — here’s what their genes can teach us

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03625-8

The representational geometry of cognitive maps under cognitive control

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-77621-x

A hybrid neural deep belief networks-gated recurrent unit algorithm with a leakage-aware evaluation protocol for ECG-HRV-based cognitive load classification

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-80282-x

Long-COVID: Ultrasensitive multiplex protein biomarker profiling reveals subclinical neuroinflammatory and innate immune dysregulation without correlation to symptom severity

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77483-9

AutoSNAP enables automated nerve segmentation and morphometry quantification

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78196-9

Information formatting in neuronal populations predicts differential functional contributions to behavior

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02514-7

Altered developmental profile of the head-direction signal in a rat model of fragile X syndrome

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11197-6

Single-neuron encoding of rapidly learned visual information reshapes human perception

Gabriel Kreiman 2026-11-04 1 min 38 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02496-6 The authors show that rapid learning modulates neuronal activity in the human visual cortex, requiring extra processing time to support recog...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02496-6

The human claustrum tracks uncertainty and prediction error

2026-11-04 1 min 54 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02471-1 Single-neuron recordings provide direct evidence that the human claustrum tracks uncertainty and prediction errors during adaptive behavior. ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02471-1

Information formatting in neuronal populations predicts differential functional contributions to behavior

Marlene R. Cohen
2026-11-10 1 min 32 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 10 November 2026; doi:10.1038/s41593-026-02514-7 Using experiments in monkeys and modeling, the authors demonstrate that the formatting of population information predicts the contribution of...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02514-7

An explainable deep learning framework for differentiating idiopathic normal pressure hydrocephalus from Alzheimer's disease using brain MRI

Sofyan Freihat
2026-11-10 1 min 269 words FRONTIERS NEUROINFORMATICS

Summary: Background and objectivesIdiopathic normal pressure hydrocephalus (NPH) and Alzheimer's disease (AD) may present with overlapping clinical and neuroimaging features, complicating differential diagnosis. This study evaluated an explainable, attention-enhanced deep learning framework for distinguishin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1990170

Perceived sound quality and speech recognition are not interchangeable outcome measures in adult cochlear implant users

Andreas Büchner 2026-11-10 1 min 198 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis study examined whether descriptor-based reports of perceived sound quality in adult cochlear implant users are associated with speech-recognition outcomes, preoperative maximum monosyllabic word recognition, and patient- or device-related factors.MethodsIn this retrospective single-...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1968712

Effects of intermittent theta burst stimulation combined with hyperbaric oxygen therapy on cognitive function and cerebral blood flow in patients with post-stroke mild cognitive impairment

Min Tang 2026-11-10 1 min 371 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo observe the effects of intermittent theta burst stimulation (iTBS) combined with hyperbaric oxygen therapy on cognitive function and cortical hemodynamics in patients with post-stroke mild cognitive impairment using functional near-infrared spectroscopy (fNIRS).MethodsSixty patients with...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966062

Assessing systematic spatial differences between intraoperative and postoperative Probabilistic Sweet Spots in Deep Brain Stimulation for Essential Tremor

Simone Hemm 2026-11-10 1 min 245 words FRONTIERS HUMAN NEUROSCIENCE

Summary: InroductionDeep Brain Stimulation (DBS) is an established therapy for movement disorders such as Essential Tremor (ET). While electrode displacement across intraoperative and postoperative stages has been studied, Probabilistic Sweet Spot (PSS) analyses rarely account for temporal evolution. Consequ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2005438

Low-intensity focused ultrasound for neuromodulation: mechanisms, parameters, and emerging clinical applications—a critical review

Yi Yang 2026-11-10 1 min 282 words FRONTIERS NEUROSCIENCE

Summary: BackgroundLow-intensity focused ultrasound (LIFU) is a non-invasive neuromodulation technique that combines millimeter-scale spatial precision with access to cortical and deep targets — including the thalamus, basal ganglia, amygdala, and nucleus accumbens — that otherwise require invasive deep brai...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2010112

Narcolepsy in the postpartum period: current evidence and future directions

Pablo Torterolo 2026-11-10 1 min 214 words FRONTIERS NEUROSCIENCE

Summary: Narcolepsy is a chronic sleep–wake disorder characterized by hypocretin (HCRT) deficiency, most commonly resulting from immune-mediated mechanisms targeting HCRT-producing neurons in the lateral hypothalamus. Although narcolepsy affects men and women at similar rates, increasing evidence suggests th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1990537

The science of neuronal regeneration: a comprehensive review on the scientific exploration of a new path of regeneration in central nervous system

Sanjay Kumar 2026-11-10 1 min 149 words FRONTIERS NEUROSCIENCE

Summary: The review focuses on neuronal regeneration in the central nervous system and the inhibition of regeneration in the CNS, such as glial scar formation, which is a crucial mechanism of extrinsic inhibition of regeneration. Regeneration-Associated Genes (RAGs), various transcription factors, epigenetic...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1951953

Shared sleep fragmentation in undergraduate roommates: dyadic actigraphy in a dense urban campus setting

Sandra E. Sephton
2026-11-10 1 min 344 words FRONTIERS NEUROSCIENCE

Summary: IntroductionDyadic sleep research has focused on romantic partners, leaving open whether nonromantic people who share a sleep space show related sleep. We examined sleep continuity and roommate concordance among undergraduate roommate dyads on a dense urban campus bordered by an interstate highway, ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1962523

Visual field performance of the novel DIMS TED spectacle lens compared with DIMS and single-vision spectacle lenses

Natalia Vlasak
2026-11-10 1 min 346 words FRONTIERS NEUROSCIENCE

Summary: PurposeTo evaluate the visual field performance of DIMS TED spectacle lens, an enhanced Defocus Incorporated Multiple Segments (DIMS) spectacle lens design featuring more defocus segments closer to the geometric center, higher defocus power and extended defocus treatment zone, compared with DIMS and...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1961854

The tau–microtubule axis in Alzheimer’s disease: reciprocal mechanisms, biomarker gaps, and therapeutic challenges

👤 Demet Özbabalık Adapınar 📅 2026-11-10 📖 1 min 📝 225 words FRONTIERS NEUROSCIENCE

Summary: Microtubule dysfunction has been implicated in axonal transport failure, synaptic degeneration, and neuronal loss in Alzheimer’s disease, yet its position within the disease process remains unresolved. This critical narrative review evaluates the tau–microtubule axis across physiological, pathologic...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2004502

Oral citicoline as add-on therapy in open-angle glaucoma: a 12-month prospective, non-randomized comparative cohort study with multimodal microperimetric, structural and OCT angiography assessment

👤 Alessandro Meduri 📅 2026-11-10 📖 1 min 📝 339 words FRONTIERS NEUROSCIENCE

Summary: BackgroundLowering intraocular pressure (IOP) is the only proven treatment for glaucoma, yet many patients progress despite adequate IOP control. Evidence for citicoline, a candidate neuroprotectant, is inconclusive and relies mainly on perimetric and electrophysiological endpoints. We evaluated the...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2004530

Maladaptive state allocation: an orexin-based framework linking sleep disturbance, chronic pain, and substance use disorder

Francisco J. Flores-Ramirez

2026-11-10 1 min 238 words FRONTIERS NEUROSCIENCE

Summary: Sleep disturbance, chronic pain, and substance use disorders are highly prevalent conditions that frequently co-occur and exacerbate one another. Although traditionally studied as distinct clinical entities, growing evidence suggests they share common neurobiological mechanisms. Among these, the ore...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2003262

Source-aware forward modelling and detection limits of acoustoelectric neural recording

Christopher Chare, Rachid Haouari, Klara Volckaert, Agknaton Bottenberg, Sebastian Haesler, Nick Van Helleputte, Xavier Rottenberg, Samer Houri and Peter Peumans

2026-11-10 1 min 298 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Acoustoelectric recording uses focused ultrasound to select the region sampled by scalp or implanted electrodes. We relate its recorded voltage to source structure, terminal conditions and the carrier phase of the focus, and identify configurations that favour neural recording. Approach. ...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aebb68

Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

👤 E Canario, C Shearer, M Akcakaya, D Weber, S Chase and J L Collinger

📅 2026-11-10 📖 1 min

📝 291 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally oc...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aebb65

Closed-loop neuroprosthetic control through spared neural activity enables proportional foot movements after spinal cord injury

👤 Vlad Cnejevici, Matthias Ponfick, Dietmar Fey, Raul C Sîmpetru and Alessandro Del Vecchio

📅 2026-11-10 📖 1 min 📝 277 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Loss of voluntary foot movement after spinal cord injury (SCI) can significantly limit independent mobility and quality of life. To improve motor output after injury, functional electrical stimulation (FES) is used to deliver stimulation pulses through the skin to affected muscles. While ...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aebb64

Golgi: an open-source graphical platform for image-to-recruitment modeling of peripheral nerve stimulation

David Lung, Yuting Jia, Roland Blumer, Lukas Reissig, Lydia M Zopf, Patrick Heimel, Christoph Kraus, Andrea Moro, Matteo Fachino and Max Haberbusch

2026-11-10 1 min 289 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Computational models of peripheral nerve stimulation—coupling finite-element bioelectric fields to biophysical axon models—are essential for designing electrodes and waveforms for neuromodulation, but established open tools are code-first, assume substantial modeling expertise, and often ...

Read full article:
https://iopscience.iop.org/article/10.1088/1741-2552/aeba17

Recent advances and challenges in autonomous wireless implantable neurologgers for animal studies: neural recording, power supply, and data transmission technologies

👤 Ye Huang, Jiaxu Cao, Xuanqi Wang, Xiaoli You, Ruiyu Bai, Mingyang Li, Yuxiang Hong, Honglong Chang, Haitao Zhao, Minghao Wang, Hao Zhang, Jun Chen and Bowen Ji

📅 2026-11-10 📖 1 min 📝 136 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Wireless implantable neurologgers are emerging as miniature autonomous systems for long-term, untethered neural recording in animal studies. However, their development remains constrained by critical engineering bottlenecks, including neural recording, power supply, and data transmission....

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb97d

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261110063344&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261110063344&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261110063344&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261110063344&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Role of smartphone-based digital adherence support for glaucoma medication therapy

👤 David J Mathew 📅 2026-11-07 📖 1 min 📝 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

Jessie Chin 2026-11-08 1 min 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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Construction of Highly Viable, Excitable, Aligned Human Induced Pluripotent Stem Cell-Derived Retinal Ganglion Cell Sheets on Poly-ε-caprolactone Nanofibers by Targeting Integrin β1-Mediated Adhesion Pathways

👤 Toru Nakazawa
📅 2026-11-09 📖 1 min 📝 63 words

LOW VISION

Summary: Transplantation of induced pluripotent stem cell-derived retinal ganglion cells (RGCs) is a promising strategy for optic nerve regeneration and vision restoration in glaucoma. However, injections of conventional single-cell suspensions have low engraftment rates, hindering the formation of the compl...

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High-Performance Self-Powered Infrared Photodetector via Semiconducting Graphene Nanoribbon/Germanium Vertical Heterojunctions with Robust Thermal Stability

👤 Aimin Song
📅 2026-11-09 📖 1 min 📝 54 words LOW VISION

Summary: Self-powered photodetectors stand as a cornerstone for next-generation low-power electronics, with promising applications in optical communications, environmental monitoring, night vision, and beyond. However, achieving high performance in these devices, particularly under elevated temperatures, rem...

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Prior-guided whole-slide survival prediction with soft histopathologic priors from vision-language foundation models

👤 Zhihui Feng
📅 2026-11-09 📖 1 min 📝 62 words LOW VISION

Summary: Whole-slide image (WSI) survival prediction is complicated by heterogeneous tumor morphology, where prognostic evidence is distributed across tumor, stroma, immune infiltration, necrosis, and residual tissue regions and often appears as mixed patch-level tissue evidence. Existing models commonly agg...

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Reduced magnocellular-biased visual responses are associated with working memory and linguistic difficulties in Chinese developmental dyslexia: A hypothesized visual-cognitive cascade

👤 Tsz Wing Leung
📅 2026-11-09 📖 1 min 📝 66 words

LOW VISION

Summary: Developmental dyslexia (DD) affects 5-10% of children worldwide. Although phonological theories predominate, they may not fully explain reading difficulties in orthographically dense scripts such as Chinese. We investigated whether Chinese children with DD exhibit differences in early visual cortica...

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During natural vision, semantic novelty modulates fixation-related processing in primate cortex

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 74 words

LOW VISION

Summary: We sample visual scenes with short gaze fixations separated by saccades. While low-level integration is known, semantic integration of foveal vision across fixations remains unclear. We hypothesized that the brain responds to changes in semantic information from one fixation to the next, and therefo...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti 📅 2026-11-05 📖 1 min 📝 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

Jianqi Li 2026-11-05 1 min 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

Sang Wook Hong 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

👤 Joseph Ollier
📅 2026-11-07 📖 1 min 📝 57 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Auditory rehabilitation in mixed hearing loss due to chronic otitis media sequelae: passive versus active transcutaneous bone conduction implant

Esther Sofía Angulo Martínez 2026-11-08 1 min 47 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: The findings of the present study suggest that AT-BCI provides a greater and clinically meaningful auditory gain for HF-PTA in patients with MHL secondary to COM. In ATBCI group a stable audiological performance was achieved as early as the 3-month follow-up and is maintained over time.

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Selective dissociation of coordination bonds enables rapid rigidity switching of hydrogel microfibre bioelectrodes

Lidong Zhang 2026-11-09 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Bioelectrodes with tunable rigidity deliver excellent mechanical compliance and can effectively alleviate tissue damage. Herein, we report hydrogel microfibre electrodes (diameter: 50-200 μm) fabricated by cross-linking sodium alginate (SA) microfibres with Ca^(2+)/Al^(3+) dual ions. The resultant S...

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Constructing Neurorights in the Era of Clinical Brain-Computer Interfaces : China's Emerging Governance Pathway

Yuan Lin 2026-11-09 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: In March 2026, China approved the world's first implantable brain-computer interface (BCI) medical device for clinical use outside clinical trials, a milestone that raises urgent questions about how legal and ethical systems should protect the brain and mind. This article examines China's emerging n...

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Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

Comparing two Python game development libraries: Pygame vs Arcade

👤 /u/Python_Data_Science 📅 2026-11-10 📖 1 min 📝 86 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I built the exact same simple mini-game using both Pygame and Arcade to see how they compare for beginners. My takeaway is that while Pygame's loop structure is great for teaching absolute programming fundamentals, Arcade is vastly superior for actual game developme...

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https://www.reddit.com/r/Python/comments/1xsho6k/comparing_two_python_game_development_libraries/

Your TV can see through walls

👤 Curiositry 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Amazon saves millions by replacing Sonnet 4.6 with Strands Decider

👤 aloukissas 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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👤 mebcitto 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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👤 werrett 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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M John Harrison becomes first science fiction author to win the Booker prize

👤 sorokod 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Fossil CO2 emissions decrease by 0.4% in 2026

👤 giuliomagnifico 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://news.exeter.ac.uk/faculty-of-environment-science-and-economy/fossil-co2-emissions-decrease-in-2026/ Comments URL: <a href="https://news.ycombinator...

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Queens Public library under ransomware attack e-borrowing suspended

👤 rustyhancock 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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AI usage generates a fragile codebase

👤 plun9 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Meet the Sad Wives of AI

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What Constant Surveillance Is Doing to Children

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> **Summary:** Article URL: https://www.theatlantic.com/magazine/2026/12/location-tracking-apps-parental-surveillance/689031/ Comments URL: <a href="https://news.ycombinator.com/item?id=5043848...

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Advancing Embodied Intelligence in Robotic-Assisted Endovascular Procedures: A Systematic Review of AI Solutions

📅 2025-12-16 📖 1 min 📝 191 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Endovascular procedures have revolutionized vascular disease treatment, yet their manual execution is challenged by the demands for high precision, operator fatigue, and radiation exposure. Robotic systems have emerged as transformative solutions to mitigate these inherent limitations. A crucial mom...

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http://ieeexplore.ieee.org/document/11301748

Charitable giving with earned and windfall money: behavioral and ERP responses to donation-risk information

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jing Wang, Xiaoya Li, Siqi Jian, Peng Li

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Transcutaneous auricular vagus nerve stimulation attenuates post-stroke dysphagia by modulating microglial polarization via the endothelial VWF/PI3K-Akt axis

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Fan Yu, Yi Feng, Yun Miao, Guangxu Xu

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https://www.sciencedirect.com/science/article/pii/S0006899326004671?dgcid=rss_sd_all

TDP-43 dysfunction destabilizes glial homeostatic identity in ALS/FTD

📖 1 min 📝 14 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Anna E Dale, Rita Sattler

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A caudate-centered network for alexithymia derived from patients with brain tumors

👤 McMahon, J. T., Healey, M. R., Sanchez-Gama, Y., Campanella, F., Skrap, M., Ferguson, M. A.

📅 2026-11-09 📖 1 min 📝 191 words BIORXIV NEUROSCIENCE

Summary: Alexithymia is a trait characterized by impaired emotional awareness. Previous accounts examining the neurobiological underpinnings of this trait have centered on a variety of structures, and alexithymia-causing lesions occur in disparate locations. We use lesion network mapping to derive a neural c...

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Linking Covariates to State Durations in Dynamic Brain Networks via Hidden Semi-Markov Models

👤 Shappell, H., Lindquist, M.

📅 2026-11-09 📖 1 min 📝 212 words BIORXIV NEUROSCIENCE

Summary: Hidden semi-Markov models (HSMMs) provide a flexible framework for studying dynamic functional connectivity because they represent brain activity as transitions between latent network states with explicit dwell time distributions. Although dwell times are often associated with behavioral and demogra...

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Electrophysiological Dynamics of Default Network Suppression

👤 Sun, W., Libowitz, M. R., Cowan, R. L., Tobel, T., Du, J., Feldman, D. A., Davis, T. S., King, J. B., Rahimpour, S., Smith, E. H., Buckner, R. L., Shofty, B.

📅 2026-11-09 📖 1 min 📝 206 words BIORXIV NEUROSCIENCE

Summary: Distributed brain networks compete with one another in the service of information processing. One dramatic example is antagonism between networks that support internally directed cognition, often collectively referred to as the default network, and networks that orient to external events and drive a...

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Sensory experience-induced plasticity of somatostatin interneurons occurs through two-pore potassium (K2P) channels

Bohannon, B. M., Alonso, N. L., Walker, M. G., Wang, M., Kaimaktchiev, K., Yang, C., Feldman, D.

2026-11-09 1 min 150 words BIORXIV NEUROSCIENCE

Summary: GABAergic interneurons (INs) are powerful modulators of plasticity and learning in cortical networks, but plasticity of INs themselves is not well understood. We studied experience-dependent plasticity in somatostatin (SST) INs, which critically regulate dendritic integration, sensory tuning, and pl...

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BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan

2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1925112

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

👤 Ansel Y. Rodríguez-González
📅 2026-10-07 📖 1 min 📝 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

👤 Bocheng Wang
📅 2026-10-07 📖 1 min 📝 217 words

FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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Multi-receptor whole-brain modeling reveals affinity-driven chemoarchitectural mechanism of virtual atomoxetine action

👤 Gustavo Deco
📅 2026-11-04 📖 1 min 📝 219 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Linking molecular mechanisms with large-scale brain dynamics remains a fundamental challenge in computational psychiatry. Receptor-informed whole-brain models have provided mechanistic insights into pharmacological interventions, yet they have largely relied on single-receptor topographies that cann...

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Editorial: Exploring the role of criticality in neural systems

👤 Tiago L. Ribeiro
📅 2026-11-04 📖 1 min 📝 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

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Music with monaural beats for anxiety reduction and mood enhancement in individuals with moderate trait-anxiety

Gianluca Micchi 2026-11-04 1 min 279 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Monaural beats embedded in music are a popular anxiety-relief intervention, yet implementation parameters and efficacy remain poorly understood. Across two studies of online participants with moderate trait anxiety, we tested whether theta-frequency (4–7 Hz) monaural beats in music produce anxiolyti...

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The association between dietary protein intake and mental disorders in older adults: a scoping review

Aliza Haslinda Hamirudin 2026-11-04 1 min 280 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMental disorders are increasingly prevalent among older adults, significantly affecting their quality of life and overall wellbeing. Recent evidence suggests that dietary factors, particularly protein intake, may play a role in mental health outcomes. However, the association between dieta...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949231

Editorial: Women in speech and language 2025

Anastasia M. Raymer 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2009604

Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

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MA-Pose and DMMS-GCN: skeleton-based martial-arts category recognition with edge-device deployment

Shilin Guo 2026-11-04 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: With the development of visual sensor technology and human pose estimation algorithms, skeleton-based action recognition has received extensive attention in sports analysis and training systems. The recognition of martial-arts actions is difficult because different styles can share similar poses whi...

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Deep reinforcement learning–driven model-based adaptive PID strategy for six-wheeled rocker–bogie mobile platform

Hongxue Feng 2026-11-04 1 min 202 words FRONTIERS NEUROROBOTICS

Summary: IntroductionSix-wheel rocker–bogie rovers encounter nonlinear dynamics, variable traction, and wheel slip that can degrade the performance of fixed-gain proportional–integral–derivative (PID) controllers. This study develops a hybrid deep reinforcement learning–PID framework to improve steering and ...

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Reliability-aware multimodal sensor fusion for human motion monitoring with safety-constrained neurorobotic feedback in sports training

Jiaoyu Wang 2026-11-04 1 min 247 words FRONTIERS NEUROROBOTICS

Summary: IntroductionEmbodied intelligent systems require reliable perception and safety-constrained feedback mechanisms for human motion understanding under heterogeneous sensing conditions. In this study, volleyball training is considered as a representative dynamic human-motion scenario rather than the so...

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The butyrate-m6A-BDNF axis: a hypothesis of gut-brain signal transduction in PSD

Dajun Gu 2026-11-04 1 min 151 words FRONTIERS NEUROSCIENCE

Summary: A reduction in brain-derived neurotrophic factor (BDNF) protein levels has been reported in post-stroke depression (PSD). However, whether transcript abundance and translation efficiency change in parallel or become uncoupled in PSD remains to be determined. In addition to classical transcriptional ...

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TEMPO-taVNS: a multimodal platform optimizing thermal, electrical, and mechanical auricular stimulation

Matthew P. Ward 2026-11-04 1 min 176 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is typically delivered electrically, while alternative stimulation modalities and their potential for personalization remain underexplored. We present TEMPO-taVNS (Thermal, Electrical, and Mechanical Parameter Optimization for taVNS), a human-...

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Baseline cognition accounts for most predictive information on cognitive recovery during rehabilitation after carbon monoxide poisoning

Peihui Liu 2026-11-04 1 min 271 words FRONTIERS NEUROSCIENCE

Summary: BackgroundCognitive change during rehabilitation after carbon monoxide (CO) poisoning may differ from prognosis at acute presentation. We examined whether information recorded in a rehabilitation pathway added predictive information beyond baseline cognition.MethodsThis single-center retrospective c...

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A purinergic receptor dominance model for microglial state transitions: balancing perturbation and restraint during neuroinflammation

Minke Tang 2026-11-04 1 min 183 words FRONTIERS NEUROSCIENCE

Summary: Microglia show substantial functional heterogeneity during neuroinflammatory responses, yet the upstream mechanisms through which they interpret extracellular perturbation-associated signals and transition between functional states remain incompletely understood. Purinergic signaling is well positio...

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Retraction: Hypothermia impairs glymphatic drainage in traumatic brain injury as assessed by dynamic contrast-enhanced MRI with intrathecal contrast

📅 2026-11-04 📖 1 min 📝 0 words FRONTIERS NEUROSCIENCE

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Delineating neural contributions to electroencephalogram-based speech decoding

👤 Motoshige Sato, Eri Hatakeyama, Masakazu Inoue, Yasuo Kabe, Sensho Nobe, Akito Yoshida, Atsushi Yamamoto, Yuya Kita, Mayumi Shimizu, Kenichi Tomeoka, Michael X Cohen and Shuntaro Sasai

📅 2026-10-30 📖 1 min 📝 249 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Speech brain–computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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Diff-ADN: a diffusion-guided artifact denoising network with deterministic residual refinement for EEG

Mudasir Jamil, Muhammad Zulkifal Aziz, Binwen Huang and Xiaojun Yu

2026-10-30 1 min

241 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Physiological artifacts degrade electroencephalographic (EEG) recordings and can affect downstream brain–computer interface (BCI) analysis. This study develops a compact single-channel framework for ocular, muscular, cardiac, and mixed-artifact removal. Approach. The Diffusion-Guided Arti...

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NeuroTest: a benchtop testbed for evaluating sensing-capable electrophysiology and neurostimulation systems

Morrow JK, H Cho, M Benjaber, Denison TJ, Widge AS and Herron JA

2026-11-04 1 min

247 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. The development of increasingly complex implantable neuromodulation devices requires extensive benchtop validation and testing prior to in vivo implementation. Our goal is to enable this in vitro testing by creating a benchtop system that is accessible, easy-to-use, and functions reliably...

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Towards semantic brain-computer interfacing: the graded inventory of semantic triggers (GIST) paradigm

👤 Joshua James Podmore, Moein Radman, Riccardo Poli, Silke Paulmann and Ian Daly

📅 2026-11-04

📖 1 min 📝 299 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Semantic brain-computer interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual p...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

Junguo Duan 2026-11-07 1 min 72 words LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

👤 Jessie Chin 📅 2026-11-08 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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Construction of Highly Viable, Excitable, Aligned Human Induced Pluripotent Stem Cell-Derived Retinal Ganglion Cell Sheets on Poly-ε-caprolactone Nanofibers by Targeting Integrin β1-Mediated Adhesion Pathways

👤 Toru Nakazawa
📅 2026-11-09 📖 1 min 📝 63 words

LOW VISION

Summary: Transplantation of induced pluripotent stem cell-derived retinal ganglion cells (RGCs) is a promising strategy for optic nerve regeneration and vision restoration in glaucoma. However, injections of conventional single-cell suspensions have low engraftment rates, hindering the formation of the compl...

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High-Performance Self-Powered Infrared Photodetector via Semiconducting Graphene Nanoribbon/Germanium Vertical Heterojunctions with Robust Thermal Stability

👤 Aimin Song
📅 2026-11-09 📖 1 min 📝 54 words LOW VISION

Summary: Self-powered photodetectors stand as a cornerstone for next-generation low-power electronics, with promising applications in optical communications, environmental monitoring, night vision, and beyond. However, achieving high performance in these devices, particularly under elevated temperatures, rem...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

👤 Ling Guan 📅 2026-11-05 📖 1 min 📝 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti
📅 2026-11-05 📖 1 min 📝 41 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

Jianqi Li 2026-11-05 1 min 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

Sang Wook Hong 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

👤 Chunxiao Wan
📅 2026-11-05 📖 1 min 📝 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

👤 Joseph Ollier
📅 2026-11-07 📖 1 min 📝 57 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Auditory rehabilitation in mixed hearing loss due to chronic otitis media sequelae: passive versus active transcutaneous bone conduction implant

Esther Sofía Angulo Martínez 2026-11-08 1 min 47 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: The findings of the present study suggest that AT-BCI provides a greater and clinically meaningful auditory gain for HF-PTA in patients with MHL secondary to COM. In ATBCI group a stable audiological performance was achieved as early as the 3-month follow-up and is maintained over time.

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Selective dissociation of coordination bonds enables rapid rigidity switching of hydrogel microfibre bioelectrodes

Lidong Zhang 2026-11-09 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Bioelectrodes with tunable rigidity deliver excellent mechanical compliance and can effectively alleviate tissue damage. Herein, we report hydrogel microfibre electrodes (diameter: 50-200 μm) fabricated by cross-linking sodium alginate (SA) microfibres with Ca^(2+)/Al^(3+) dual ions. The resultant S...

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Constructing Neurorights in the Era of Clinical Brain-Computer Interfaces : China's Emerging Governance Pathway

Yuan Lin 2026-11-09 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: In March 2026, China approved the world's first implantable brain-computer interface (BCI) medical device for clinical use outside clinical trials, a milestone that raises urgent questions about how legal and ethical systems should protect the brain and mind. This article examines China's emerging n...

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

2024-01-22 1 min 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition’s foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Tuesday Daily Thread: Advanced questions

👤 /u/ AutoModerator 📅 2026-11-10 📖 1 min 📝 288 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Wednesday Thread: Advanced Questions 🐍</h1> Dive deep into Python with our Advanced Questions thread! This space is reserved for questions about more advanced Python topics, frameworks, and best practices. <h2>How it Works:</h2> Ask A...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xs81uc/tuesday_daily_thread_advanced_questions/

Show HN: Someone Was Here – Find out who used your Mac while you were away

👤 4onthefloor124 📅 2026-11-09 📖 1 min 📝 114 words HACKER NEWS

> **Summary:** Many thanks to my ex for inspiring my new MacOS app, Someone Was Here. It takes a photo of whoever uses your Mac while you're away, whether you left it unlocked or they know your password, and sends their photo to your iPhone if the face isn't yours. I built Someone Was Here as a menu bar app...

🔗 **Read full article:**
https://someonewashere.app/

Parking-app convenience fees in 312 U.S. cities (median $0.35 per session)

👤 parkgraph 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://parkgraph.com/parking-app-fees Comments URL: https://news.ycombinator.com/item?id=50430991 Points: 62 # Comments: 39

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https://parkgraph.com/parking-app-fees

TPU 8tx and TPU 8i Max

👤 _benx_ 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.google/innovation-and-ai/infrastructure-and-cloud/global-network/tpu-8i-max-and-8tx/ Comments URL: <a href="https://news.ycombinator.com/item?id=5043283...

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Exact – a new cross platform UI framework

👤 vmg12 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/expo/exact Comments URL: https://news.ycombinator.com/item?id=50433541 Points: 64 # Comments: 30

🔗 **Read full article:**
https://github.com/expo/exact

Fable 5.2 one-shotted my nephew's dream game

👤 lexandstuff 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://notesbylex.com/fable-5-2-one-shotted-my-nephews-dream-game Comments URL: https://news.ycombinator.com/item?id=50433659 Po...

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Horible public key tricks

👤 Curiositry 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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https://maurycyz.com/projects/badcert/

Locating the source of a mysterious beep

👤 bigwheels 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://simonwillison.net/2026/Nov/9/mysterious-beeps/ Comments URL: https://news.ycombinator.com/item?id=50434310 Points: 30 # Commen...

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https://simonwillison.net/2026/Nov/9/mysterious-beeps/

New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

> **Summary:** We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

Summary: In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

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Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

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Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

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Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

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VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

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http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

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A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

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Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

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http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

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Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Table of Contents

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Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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http://ieeexplore.ieee.org/document/11364153

Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Table of Contents

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Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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http://ieeexplore.ieee.org/document/11272135

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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http://ieeexplore.ieee.org/document/11743154

Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

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Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

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EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

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Repetition dissociates pointer and content-based representations in visual working memory: contrasting the CDA with multivariate shape decoding

📖 1 min 📝 16 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Dock H. Duncan, Güven Kandemir, Christian N.L. Olivers

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006506?dgcid=rss_sd_all

Distinguishing evidence accumulation from choice confidence in neural signatures of perceptual decision making

👤 Monakhovych, D., Grogan, J. P., Kelly, S. P., O'Connell, R. G. 📅 2026-11-09 📖 1 min 📝 235 words

BIORXIV NEUROSCIENCE

> **Summary:** Human electrophysiology research has identified the centro-parietal positivity (CPP) as an apparent neural correlate of sensory evidence accumulation during perceptual decision-making. The CPP has also been shown to closely scale with subjective confidence. Coupled with the fact of its domain-genera...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.29.760356v1?rss=1

Mapping and longitudinal imaging of potentiated synapses in the mouse hippocampus using SynActive-eGRASP

👤 Latini, F. C., Jacob, A., Mirzapourdelavar, H., Cattaneo, A., Attardo, A., Mainardi, M.

📅 2026-11-09

📖 1 min 📝 185 words

BIORXIV NEUROSCIENCE

Summary: Identifying individual synapses that undergo long-term potentiation following learning at specific circuits remains a major challenge in neuroscience. To address this problem, we developed SA-eGRASP, a genetically encoded reporter that combines SynActive (SA) targeting with enhanced GFP Reconstituti...

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A Neurocompliant Ionic Conductive Hydrogel Orchestrates Glial and Vascular Remodeling to Mitigate Traumatic Brain Injury

Okoro, P. D., Jaberi, R., Blackburn, S., Patel, D., Halasah, N., Mapa, J., Shih, M., Botzbach, R., Tan, J., Vullev, V. I., Liu, H., Santhakumar, V., Noshadi, I.

2026-11-09 1 min 142 words BIORXIV NEUROSCIENCE

Summary: Developing conductive scaffolds for neural regeneration is often hindered by the cytotoxicity and mechanical mismatch inherent to conventional electronic fillers. Here, we report an ultrasoft ionically conductive hydrogel engineered by grafting choline-based bioionic liquids (BIL) onto fish gelatin ...

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https://www.biorxiv.org/content/10.64898/2026.10.29.762869v1?rss=1

Ergodic breakdown of brain dynamics under chronnectome complexity across cognitive states and age

Kazemivash, B., Zhu, D.

2026-11-09 1 min 289 words BIORXIV NEUROSCIENCE

Summary: Chronnectome complexity, the extent to which time-varying functional connectivity follows a structured sequence of distinct configurations, contributes to shaping large-scale brain dynamics. However, it remains unclear how this complexity influences the encoding of information that distinguishes res...

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https://www.biorxiv.org/content/10.64898/2026.11.06.764964v1?rss=1

Mitochondrial Dysfunction is Part of a Novel Mechanism for Chemotherapy-Induced Neuropathic Pain

👤 Sakas, R., Abu-Ata, S., Nakhleh-Francis, Y., Abd Elkarim, N., Kotiyal, A., Politte, H., Awad-Igbaria, Y., F. Soustiel, J., Palzur, E.

📅 2026-11-09 📖 1 min 📝 213 words BIORXIV NEUROSCIENCE

Summary: Chemotherapy-induced peripheral neuropathy (CIPN) is a common, persistent complication of cancer treatment, yet the mechanisms sustaining pain remain unclear. Mitochondrial dysfunction has been implicated in chronic pain, but its role in spinal cord dysfunction after chemotherapy, and whether it can...

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Identifying a neurochemical signature of the paraventricular nucleus of the thalamus that predicts individual variability in cue-reward learning

👤 Flagel, S. B., Ramos, B. A., Chakrabarty, S., Cain, C. N., Levina, E., Kennedy, R. T.

📅 2026-11-09

📖 1 min 📝 228 words

BIORXIV NEUROSCIENCE

Summary: To investigate neurochemical factors that may confer vulnerability to psychiatric disorders, we used an animal model that captures a transdiagnostically relevant neurobehavioral endophenotype. During a Pavlovian conditioned approach paradigm, rats learn the predictive relationship between a lever cu...

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https://www.biorxiv.org/content/10.64898/2026.11.06.764855v1?rss=1

Allele-agnostic KCNH1 suppression rescues gain-of-function developmental epileptic encephalopathy

Chi, W., Corbett, E. E., Thompson, C. H., Simkin, D., Adil, H., Hawkins, N. A., Callow, N. A., Barse, L., George, A. L., Kearney, J. A., Kiskinis, E.

2026-11-09 1 min 214 words BIORXIV NEUROSCIENCE

Summary: Gain-of-function variants in KCNH1, which encodes the voltage-gated potassium channel Kv10.1, cause developmental and epileptic encephalopathies (DEEs) characterized by seizures, developmental delay and intellectual disability. While pathogenic variants enhance channel activity, loss of KCNH1 is tol...

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Allele-selective knockdown of mutant DNAJB6 improves molecular features in a primary mouse myotube model of LGMDD1

Vohra, A., Haller, M., Paing, M. M., Dam, A. B. L., Daw, J. A., Bengoechea, R., Pittman, S. K., Wang, T.-Y., Yuede, C. M., Miller, T. M., True, H. L., Harper, S. Q., Chou, T.-F., Weihl, C. C., Findlay, A. R.

2026-11-09 1 min 197 words BIORXIV NEUROSCIENCE

Summary: Dominant variants in DNAJB6 cause limb-girdle muscular dystrophy D1 (LGMDD1), a progressive myopathy associated with disrupted protein homeostasis. Because complete loss of DNAJB6 is embryonic lethal in mice, we evaluated allele-selective knockdown as a strategy to reduce mutant DNAJB6 while preserv...

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Variable errors suppress implicit motor adaptation among older adults

Zhou, Y., Andrews, I., Sang, A. F., Torres, J. M., Kim, O. A.

2026-11-09 1 min 152 words BIORXIV NEUROSCIENCE

Summary: Age-related changes in motor control networks manifest in a highly task-dependent manner. Although older adults can display enhanced implicit sensorimotor adaptation, this effect is inconsistent across perturbation types (e.g., visuomotor vs. force-field perturbations). Because these perturbations i...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.06.764937v1?rss=1

Comparing apples to oranges from photoreceptors to inferior temporal cortex through neural manifolds

👤 Xu, Y. G., Maruya, A., Conway, B., Zaidi, Q.
📅 2026-11-09 📖 1 min 📝 100 words

BIORXIV NEUROSCIENCE

Summary: In the neural processing of color, cone photoreceptor outputs are combined over successive stages to form inferior temporal cortex (IT) cells with narrower color tuning and more diverse peak preferences. To understand the benefits of these transformations, we explored the capability of photoreceptor...

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https://www.biorxiv.org/content/10.64898/2026.10.30.763101v1?rss=1

The effects of smartphone presence during a reading task: a psychophysiological study

📅 2026-11-09 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79191-w

Intracortical recordings in human lateral occipital complex support the biased-competition model of spatial attention

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-78323-6

MER-CycleGAN: memory-embedded residual learning for brain MRI synthesis and harmonization

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High school science ambitions

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41592-026-03323-3

Long-term development of a motor memory

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41562-026-02622-1

Saudi Arabic automatic speech recognition and dialect identification with calibrated system combination and acoustic–textual modeling

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79593-w

Increased functional connectivity in glioma: separating compensation from tumor-promoting remodeling

Ning Su 2026-11-09 1 min 134 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Increased functional connectivity (FC) in glioma is often interpreted as compensatory neuroplasticity or tumor-promoting circuit remodeling, although FC is a measure of coupling rather than a mechanism. We propose a qualitative framework that first examines whether the measurement is suitable for th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2005308

Context-dependent brain connectivity reorganization following chronic antibiotic exposure during social behavior

Joseph Carrión 2026-11-09 1 min 179 words FRONTIERS NEUROSCIENCE

Summary: IntroductionThe gut microbiome influences brain function, but how chronic antibiotic exposure affects neural network organization during social behavior remains unclear. We investigated the effects of microbiome disruption on brain metabolism and functional connectivity.MethodsMale C57BL/6J mice rec...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1936717

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

👤 Junguo Duan
📅 2026-11-07 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

👤 Jessie Chin 📅 2026-11-08 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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Construction of Highly Viable, Excitable, Aligned Human Induced Pluripotent Stem Cell-Derived Retinal Ganglion Cell Sheets on Poly-ε-caprolactone Nanofibers by Targeting Integrin β1-Mediated Adhesion Pathways

👤 Toru Nakazawa
📅 2026-11-09 📖 1 min 📝 63 words

LOW VISION

Summary: Transplantation of induced pluripotent stem cell-derived retinal ganglion cells (RGCs) is a promising strategy for optic nerve regeneration and vision restoration in glaucoma. However, injections of conventional single-cell suspensions have low engraftment rates, hindering the formation of the compl...

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High-Performance Self-Powered Infrared Photodetector via Semiconducting Graphene Nanoribbon/Germanium Vertical Heterojunctions with Robust Thermal Stability

👤 Aimin Song
📅 2026-11-09 📖 1 min 📝 54 words LOW VISION

Summary: Self-powered photodetectors stand as a cornerstone for next-generation low-power electronics, with promising applications in optical communications, environmental monitoring, night vision, and beyond. However, achieving high performance in these devices, particularly under elevated temperatures, rem...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti
📅 2026-11-05 📖 1 min 📝 41 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li
📅 2026-11-06 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier
2026-11-07 1 min 57 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Auditory rehabilitation in mixed hearing loss due to chronic otitis media sequelae: passive versus active transcutaneous bone conduction implant

Esther Sofía Angulo Martínez 2026-11-08 1 min 47 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: The findings of the present study suggest that AT-BCI provides a greater and clinically meaningful auditory gain for HF-PTA in patients with MHL secondary to COM. In ATBCI group a stable audiological performance was achieved as early as the 3-month follow-up and is maintained over time.

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Selective dissociation of coordination bonds enables rapid rigidity switching of hydrogel microfibre bioelectrodes

Lidong Zhang 2026-11-09 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Bioelectrodes with tunable rigidity deliver excellent mechanical compliance and can effectively alleviate tissue damage. Herein, we report hydrogel microfibre electrodes (diameter: 50-200 μm) fabricated by cross-linking sodium alginate (SA) microfibres with Ca^(2+)/Al^(3+) dual ions. The resultant S...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

📅 2026-07-30 📖 1 min 📝 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

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http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

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http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

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http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001096

Python events between Nov 09 - Nov 18

👤 /u/szabgab 📅 2026-11-09 📖 2 min 📝 464 words REDDIT PYTHON

Summary: <!-- SC_OFF --> In the age of AI I think personal connections are even more important than before. That's why I try to help promoting events where people can meet in-person or virtually and discuss the topics that interest them. I found the following Python-related events fo...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xrsoay/python_events_between_nov_09_nov_18/

Infineon marks innovation milestone with 300mm power silicon carbide wafer

👤 sshah03 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.infineon.com/press-release/2026/infpr202611-028 Comments URL: https://news.ycombinator.com/item?id=50429430 Points: 54 <...

🔗 **Read full article:**
https://www.infineon.com/press-release/2026/infpr202611-028

Too many men suffering in silence, national health report finds (Canada)

👤 nosecreek 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.westcentralonline.com/articles/too-many-men-suffering-in-silence-national-health-report-finds Comments URL: <a href="https://news.ycombinator.com...

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https://www.westcentralonline.com/articles/too-many-men-suffering-in-silence-national-health-report-finds

DeepX claims inference up to 100B parameters at less than 5 W

👤 giuliomagnifico 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.eetimes.com/deepx-tapes-out-second-gen-chip-projects-6-7x-performance-gain-with-lpddr5x-pim/ Comments URL: <a href="https://news.ycombinator.com/i...

🔗 **Read full article:**
https://www.eetimes.com/deepx-tapes-out-second-gen-chip-projects-6-7x-performance-gain-with-lpddr5x-pim/

Show HN: MeowDisplay Is iPhone Mirroring, but in Reverse

👤 raiseCatError 📅 2026-11-09 📖 1 min 📝 288 words HACKER NEWS

> **Summary:** Hey HN and Cats, thank you for checking this out! Github: https://github.com/raiseCatError/meowdisplay EDIT 2: I should've mentioned that MeowDisplay started as a fork of OpenDisplay: <a href="https://github.com/peetzwe...

🔗 **Read full article:**
https://meowdisplay.app/

E-means: 10x faster k-means at equal loss

👤 porridgeraisin 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://e-means.github.io/ Comments URL: https://news.ycombinator.com/item?id=50429734 Points: 88 # Comments: 3

🔗 **Read full article:**
https://e-means.github.io/

Claude Fable 5.2 and Mythos 5.2

👤 meetpateltech 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.anthropic.com/claude-fable-and-mythos-5-2 Comments URL: https://news.ycombinator.com/item?id=50429906 Points: 596 # Comment...

🔗 **Read full article:**
https://www.anthropic.com/claude-fable-and-mythos-5-2

Iran was behind cyber-attack on French supermarket freezers, Paris says

👤 disenchanted 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theguardian.com/world/2026/nov/09/france-blames-irans-revolutionary-guards-for-cyber-attacks-on-supermarket-fridges Comment...

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CUI – Native UI components for C

👤 hilti 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://yappologistic.github.io/CUI/ Comments URL: https://news.ycombinator.com/item?id=50430621 Points: 54 # Comments: 16

🔗 **Read full article:**
https://yappologistic.github.io/CUI/

USPS now offers official ZIP Code boundary polygons

👤 munro 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://postalpro.usps.com/zcb Comments URL: https://news.ycombinator.com/item?id=50431058 Points: 57 # Comments: 26

🔗 **Read full article:**
https://postalpro.usps.com/zcb

Claude Fable 5.2 Scores 65 on Artificial Analysis, Highest So Far

👤 theanonymousone 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://artificialanalysis.ai/models/claude-fable-5-2 Comments URL: https://news.ycombinator.com/item?id=50431144 Points: 11 # Comments...

🔗 **Read full article:**
https://artificialanalysis.ai/models/claude-fable-5-2

The Relational Capacity Framework: A Lens for AI-Mediated Relationality

Emily Tseng, Patrick Gage Kelley, Renee Shelby, Ashley M. Walker

2026-11-09 1 min

154 words

ARXIV CS HC

Summary: arXiv:2611.08877v1 Announce Type: new Abstract: A growing literature examines the impact of engaging with AI chatbots on our relationality: our ability to form relationships with others, maintain them through life's changes, and exit when needed. Currently, substantial work focuses on purpose-built...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08877

Bidirectional Modeling of mmWave Human Signals for Enhancement & Synthesis

Shuntian Zheng, Hoang Hai Pham, Jiaqi Li, Xiaoman Lu, Shuai He, Yu Guan

2026-11-09 1 min

259 words

ARXIV CS HC

Summary: arXiv:2611.08870v1 Announce Type: new Abstract: Millimeter-wave (mmWave) is attractive for human-centered robotics in smoke, darkness, and privacy-sensitive indoor spaces: it measures reflections rather than visually identifiable imagery and remains usable without light. Deployed systems, however, ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08870

"I had nobody to turn to": Why LGBTQ+ People Choose (Not) to Use AI Chatbots for Personal Health and Relationship Advice

👤 Jacob Hobbs, Ashray Gupta, Dennis Ulmer, Geetanjali Aich, Nolan Golden, Yanan Long, Ian Rios-Sialer, Jennifer Mickel, Morgan Klaus Scheuerman, Rohit Avinash Mujumdar, Virginia K. Felkner, Jordan Taylor, William Agnew

📅 2026-11-09 📖 1 min 📝 207 words ARXIV CS HC

Summary: arXiv:2611.08854v1 Announce Type: new Abstract: While there is an abundance of research into large language model (LLM) biases against LGBTQ+ communities, far less research has examined how and why LGBTQ+ people choose to make use of AI chatbots. In this work, we examine how LGBTQ+ people choose (n...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08854

Lea: An Agent Backbone for Mathematician-Controlled Formalization in Lean

Shaswat Patel, Daniel Aarao Reis Arturi, Kevin Gomez, Jaume de Dios Pont, Jorge Poco, Eduardo V. Teixeira, Sam Westrick, Claudio Silva, Chinmay Hegde

2026-11-09 1 min 200 words ARXIV CS HC

Summary: arXiv:2611.08849v1 Announce Type: new Abstract: Agentic systems now formalize research-level mathematics in Lean, but they are built for autonomy: the mathematician states a target and inspects a result. The alternative, a general-purpose coding harness, is domain-neutral and must be assembled into...

Read full article:
https://arxiv.org/abs/2611.08849

Driver Takeovers and Longitudinal Responses Analysis from Naturalistic Driving

Yuhang Wang, Hao Zhou

2026-11-09 1 min 276 words ARXIV CS HC

Summary: arXiv:2611.08798v1 Announce Type: new Abstract: As Level 2 advanced driver-assistance systems (ADAS) become more widely used, understanding driver-system interaction is increasingly important for traffic safety. However, limited real-world data make these interactions difficult to study. We present...

Read full article:
https://arxiv.org/abs/2611.08798

MotionDecipher: LLM-Assisted Clinical Interpretation of Human Motion Signals for Efficient Gait Analysis

👤 Mingrui Ma, Tian Tan, Jinsi Guo, Jessica Rose, Yiwen Dong

📅 2026-11-09 📖 1 min 📝 204 words

ARXIV CS HC

Summary: arXiv:2611.08795v1 Announce Type: new Abstract: Clinical gait and motion analysis is widely used to guide targeted interventions for patients with neurological disorders, the outcome of which is typically a report with descriptions of joint movements and forces. Current practices rely on experts' m...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08795

Learning from How Users Interpret: Personalizing Recommendation Explanation Order

👤 Deo Munduku, Elsa Negre

📅 2026-11-09 📖 1 min 📝 194 words

ARXIV CS HC

Summary: arXiv:2611.08785v1 Announce Type: new Abstract: Personalized explanations typically adapt their content or presentation based on preferences, interaction histories, or other user characteristics. We examine whether regularities inferred from prior textual interpretations can support personalization...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08785

Goals as a Representational Layer for Human-AI Collaboration

Gaia Molinaro, Lev Tankelevitch, Sean Rintel, Scott Counts

2026-11-09 1 min 149 words

ARXIV CS HC

Summary: arXiv:2611.08757v1 Announce Type: new Abstract: Goals are central to how people, teams, and organizations structure their work. Yet, while generative artificial intelligence (AI) systems increasingly operate across a worker's context, information about why a person is pursuing a task, beyond local ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08757

Seeing Is Not Knowing: The Epistemic Limits of LLM-Based Mental Health Inference from Youth Social Media Conversations

Xiaohan Ding, Mengting Tu, Andrew H. Zhao, Jamie Zelazny, Candice L. Biernesser, Munmun De Choudhury

2026-11-09 1 min 152 words

ARXIV CS HC

Summary: arXiv:2611.08565v1 Announce Type: new Abstract: Adolescents increasingly use social media for private conversations, sharing experiences and seeking support. These interactions generate digital traces, prompting efforts to infer mental health and psychosocial states. As large language models become...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08565

Relic and Gumi: Governing Memory, Safety, and Continuity in Diegetic Relational Agents

Daniele Veri'
2026-11-09
1 min
231 words
ARXIV CS HC

Summary: arXiv:2611.08504v1 Announce Type: new Abstract: Longitudinal relational agents rely on memory for continuity, but remembering can lead to intrusion, clinical overconfidence, emotional manipulation, and opacity. We present Relic, a governance architecture for diegetic relational agents, and Gumi, a ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08504

Oscillatory Phase Coding Emerges in Adaptive RNNs for Flexible Working Memory

Kehan Lu, Tianhao Chu, Aiersi Tuerhong, Yiwen Sun, Huan Luo, Fang Fang, Si Wu
2026-11-09
1 min
215 words
ARXIV QBIO NC

Summary: arXiv:2611.06932v2 Announce Type: replace Abstract: Oscillatory phase coding (OPC) is widely observed in neural systems and has been implicated in cognitive functions such as memory and sequence processing. However, its computational role and neural substrates remain largely unclear. Here, we studi...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06932

Multimodal Physiological Decoding Reveals Individualized Arousal Dynamics in Closed-Loop Neurofeedback

👤 Anirudh Natarajan, Paul Sajda
📅 2026-11-09 📖 1 min 📝 270 words

ARXIV QBIO NC

Summary: arXiv:2610.04113v2 Announce Type: replace Abstract: Most arousal-based brain-computer interfaces (BCIs) use EEG to decode cognitive state. But arousal is an autonomic process. We examined if peripheral physiological signals give a better decode of task-related arousal. We used a public dataset from...

🔗 **Read full article:**
https://arxiv.org/abs/2610.04113

Platonic Representations in the Human Brain: Unsupervised Recovery of Universal Geometry

👤 Pablo Marcos-Manch\'on, Rishi Jha, Llu\'is Fuentemilla
📅 2026-11-09 📖 1 min 📝 139 words

ARXIV QBIO NC

Summary: arXiv:2605.20496v2 Announce Type: replace Abstract: The Strong Platonic Representation Hypothesis suggests that representational convergence in artificial neural networks can be harnessed constructively: embeddings can be translated across models through a universal latent space without paired data...

🔗 **Read full article:**
https://arxiv.org/abs/2605.20496

A multi-scale information geometry reveals the structure of mutual information in neural populations

Simone Azeglio, Steeve Laquitaine, Ulisse Ferrari, Matthew Chalk

2026-11-09 1 min

204 words

ARXIV QBIO NC

Summary: arXiv:2605.06304v2 Announce Type: replace Abstract: Understanding how neural population responses represent sensory information is a central problem in systems neuroscience. One approach is to define a representational geometry on stimulus space in which distances reflect how reliably stimuli can b...

🔗 **Read full article:**
https://arxiv.org/abs/2605.06304

Traits and tangles: An analysis of the Big Five paradigm by tangle-based clustering

Hanno von Bergen, Reinhard Diestel

2026-11-09 1 min 227 words

ARXIV QBIO NC

Summary: arXiv:2411.18670v2 Announce Type: replace Abstract: Using the recently developed mathematical theory of tangles, we re-assess the mathematical foundations for applications of the five factor model in personality tests by a new, mathematically rigorous, quantitative method. Our findings broadly conf...

🔗 **Read full article:**
https://arxiv.org/abs/2411.18670

A hair-thin implant opens the entire depth of the mouse brain to revisiting the same structures at subcellular resolution

👤 Miroslav Stib\r{u}rek, Tom\'a\v{s} Pik\'alek, Petra Ondr\'a\v{c}kov\'a, Kristina Herz\'anov\'a, Adrian Lorenz, Andr\'e Gomes, Julia Henschke, Endre Levente Marosi, Bhargavi Murthy, Petra Kolb\'abkov\'a, Hongbo Jia, Sanja Bauer Mikulovic, Janelle Pakan, Alessio Attardo, Hana Uhl\'i\v{r}ov\'a, Tom\'a\v{s} \v{C}i\v{z}m\'ar

📅 2026-11-09 📖 1 min 📝 137 words ARXIV QBIO NC

Summary: arXiv:2611.09481v1 Announce Type: cross Abstract: Ultrathin multimode-fibre endoscopes provide minimally invasive access to subcellular structures throughout the depth of the mouse brain. Here we introduce a chronically implantable interface with a diameter of 144 {\mu}m, about the thickness of a h...

🔗 **Read full article:**
https://arxiv.org/abs/2611.09481

Decoding spatial and semantic information from large-scale visual cortical activity for natural-movie reconstruction

Yukai Huang, Haoyu Albert Wang, Jiawei Zhang, Jialun Ma, Jiayao Ding, Yiran Lin, Xingyu Chen, Ziqi Gao, Yuguo Yu

2026-11-09 1 min 189 words ARXIV QBIO NC

Summary: arXiv:2611.08887v1 Announce Type: cross Abstract: Reconstructing natural visual experience from cortical population activity is a central challenge in neural decoding, particularly for calcium imaging, where signals are noisy, temporally filtered, and paired neural-video data are limited. Here we p...

Read full article:
https://arxiv.org/abs/2611.08887

Path-dependent triage: Early judgments lead conversational AI chatbots to miss mental-health emergencies

Veith Weilnhammer, Raymond Dolan, Lennart Luettgau, Virginia Corno, Elise Wilkinson, Viknesh Sounderajah, Hannah Richardson, Christopher Kelly, Matthew M Nour

2026-11-09 1 min 273 words ARXIV QBIO NC

Summary: arXiv:2611.09647v1 Announce Type: new Abstract: People increasingly bring mental-health concerns to consumer AI chatbots. In these conversations, clinically relevant information rarely arrives all at once. Instead, it unfolds over multiple turns, often in response to the questions the chatbot asks....

Read full article:
https://arxiv.org/abs/2611.09647

Spiking neuromodulation: a linear control perspective

Fulvio Forni, Rodolphe Sepulchre
2026-11-09 1 min 82 words ARXIV QBIO NC

Summary: arXiv:2611.08571v1 Announce Type: new Abstract: Neuronal excitability is a nonlinear phenomenon, and the complexity of spiking experimental traces sometimes looks defeating. The present paper takes the view point that excitability is a neuromodulated property, and aims to demonstrate that the neuro...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08571

Structure is in the zoom: probing neural symmetry through dimensionality scaling

Adil Amin
2026-11-09 1 min 308 words ARXIV QBIO NC

Summary: arXiv:2611.08506v1 Announce Type: new Abstract: How much of a neural population's measured dimensionality is the code, and how much is the act of measuring it? The slope of the participation ratio against subsample size is reported for cortical populations and for the layers of language models as a...

🔗 **Read full article:**
https://arxiv.org/abs/2611.08506

Plasma p-tau181, Aβ1-42, and GFAP across Alzheimer’s disease, mixed-type Alzheimer’s disease, mild cognitive impairment, anxiety disorders, and depressive disorders

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Haifeng Lou, Chenyun Huang, Ji Shen, Dake Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004683?dgcid=rss_sd_all

RAD51 expression in pan-cancer and its prognostic and immune mechanisms in glioma

📖 1 min 📝 35 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Liubing Hou, Yu Wang, Xiang Zhan, Jiayuan Li, Zizhou Zhang, Weijing Cai, Linlin Su, Wei Han, Xuetao Han, Yue Han, Xiaolan Lv, Xiaoying Xue, Huandi Zhou

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004816?dgcid=rss_sd_all

Reduced hypothalamic [³H]PK11195 binding in Lewis rats after chronic high-intensity interval training

📖 1 min 📝 28 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Atiqur Hasany, John R. Tyler, Huy Lu, Kenneth Blum, Wei Sun, Richard J. Salvi, Igor Elman, Panayotis K. Thanos

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004622?dgcid=rss_sd_all

Plasma miR-21-5p, miR-195-5p, and clusterin profiles in relapsing-remitting multiple sclerosis and their relationship with cognitive function

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Osman Oğuzhan Kılıç, Semra Mungan, Birsen Can Demirdöğen

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006846?dgcid=rss_sd_all

Obesity-induced oxidative stress and cGMP depletion suppress cerebellar α2,3-Na⁺,K⁺-ATPase activity in female mice

📖 1 min 📝 61 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Martina Raissa Ribeiro, Geovanni de Morais Lima, Amanda Siena, Jacqueline Alves Leite, Victoria Francelino Alves, Vitoria Cristina Araujo, André Schmidt Suaiden, Bruno de Souza Gonçalves, Marina Marq...

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Clip-to-trial: EEG-based decoding network with long short-term self-attention for DRDS recognition

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Lili Shen, Yiming Wu, Zihao Yang, Xinyuan Di

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007074?dgcid=rss_sd_all

Intranasal insulin improves long-term cognitive deficits and protects hippocampal damage following neonatal hypoxia–ischemia in juvenile rats

📖 1 min 📝 41 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Nilesh Dankhara, Jonathan W. Lee, Norma B. Ojeda, Michelle A. Tucci, Silu Lu, Elizabeth L. White, Lindsey E. Winborne, Sharon Xu, Carolyn K. Glendye, Lu-Tai Tien, Yi Pang, Abhay J. Bhatt, Lir-Wan Fan</...

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https://www.sciencedirect.com/science/article/pii/S0306452226007153?dgcid=rss_sd_all

Characterization of the hemodynamic response function in default mode network subregions for task-evoked negative BOLD responses

📖 1 min 📝 29 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Hengda He, Bardiya G. Yazdi, Amirreza Sedaghat, Seyed Hani Hojjati, Sindy Ozoria, Peter Chernek, Jenseric Calimag, Farnia Feiz, Qolamreza R. Razlighi

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006336?dgcid=rss_sd_all

A multimodal MRI study of DTI-ALPS and cerebral blood flow changes across the Alzheimer's disease continuum

📖 1 min 📝 38 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yian Gao, Manxi Xu, Peiling Ou, Haoyu Yang, Jia Wang, Wulong Liu, Yu Sun, Wenying Du, Yuli Wang, Tongxi Liu, Hui Xia, Liping Fu, Ni Shu, Zhen Yuan, Guolin Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006373?dgcid=rss_sd_all

Targeted somatosensory stimulation modulates facial nucleus glial responses and promotes functional recovery after facial nerve axotomy in rats

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-80244-3

Modulation of cell death and neurogenesis by early intervention with sodium butyrate and galantamine in a streptozotocin-induced Alzheimer’s rat model

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77208-y

Association between preoperative bisphosphonate therapy and postoperative neurocognitive-related outcomes in osteoporosis: a nationwide propensity score–matched cohort study

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-80129-5

Calpain-mediated TBP cleavage disturbs neuronal AMOT-YAP1 signalling

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79055-x

Hierarchical modality reconstruction and transformer interaction with adaptive fusion for multimodal sentiment analysis with missing modalities

📅 2026-11-09 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78483-5

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

👤 Yingjie Zhu 📅 2026-11-03 📖 1 min 📝 24 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 03 November 2026; doi:10.1038/s41593-026-02535-2 Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02535-2

Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

Ole A. Andreassen | 2026-11-06 | 1 min | 25 words | NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 06 November 2026; doi:10.1038/s41593-026-02536-1 Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02536-1

Neuroimmune-endocrine interactions in perinatal asphyxia: from brain injury to neuroprotection

Francisco Capani | 2026-11-06 | 1 min | 295 words | FRONTIERS HUMAN NEUROSCIENCE

Summary: Perinatal asphyxia (PA) remains a major cause of neonatal mortality and long-term neurodevelopmental disability worldwide. Although hypoxic–ischemic brain injury has traditionally been attributed to oxygen deprivation, excitotoxicity, and oxidative stress, accumulating evidence indicates that neurol...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2018788

Meditation as a bioactive intervention: evidence of translational potential via Connectivity Mapping and TMS neurophysiology

Mitali Mukerji 2026-11-09 1 min 295 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMeditation practices are often used as non-pharmacological adjunct therapy for managing stress and wellbeing benefits. Here, we investigate whether molecular signatures via meditation practices have potential overlap with those of known drug targets that have established neurophysiological...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1904068

Hearing loss as an inference problem: sensory precision, conversational repair, and partner adaptation

Sebastian Schnettler 2026-11-09 1 min 289 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conversational understanding is often approached as a problem of successful signal transmission. This paper develops a complementary account in which conversation is treated as a reciprocal process of inference and action: interlocutors must infer the latent causes of one another’s signals, estimate...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1963610

Noise resilience in brain decoding driven by spiking neural networks for responsive robot control

👤 Alessandro De Luca 📅 2026-11-09 📖 1 min 📝 291 words FRONTIERS NEUROROBOTICS

Summary: Brain computer interface (BCI) is a promising technology, expanding on many fields ranging from medical to commercial applications. The diffusion of this technique is compromised by the complexity of the treated data and the unavoidable presence of noise in those systems. This study contributes to t...

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A deep-learning microglial response score integrates single-cell and spatial transcriptomics for Alzheimer’s disease stratification

👤 Li Ling 📅 2026-11-09 📖 1 min 📝 270 words FRONTIERS NEUROSCIENCE

Summary: BackgroundMicroglia exhibit diverse responses in Alzheimer’s disease (AD), yet linking these responses to tissue pathology and disease heterogeneity remains challenging. We aimed to develop a deep-learning microglial response score (DL-MRS) integrating single-cell, spatial and bulk transcriptomics t...

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Perceptual versus memory-guided affordances: training-related adaptations in a spatial computer game

Alicja M. Olszewska 2026-11-09 1 min 205 words FRONTIERS NEUROSCIENCE

Summary: Traditional sequence learning research relies on discrete button-press paradigms where pauses segment responses. We developed a continuous game-based adaptation of the Serial Reaction Time task and tracked neural changes in 25 participants using fMRI before and after 1–3 weeks of training. Participa...

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Early social enrichment enhances adult social investigation and increases oxytocin and vasopressin receptor mRNA expression in the nucleus accumbens of adult female mice

Rossella Ventura 2026-11-09 1 min 197 words FRONTIERS NEUROSCIENCE

Summary: IntroductionThe early postnatal period is a sensitive window for the development of social behavior, during which interactions with caregivers play a critical role in shaping brain circuits and neuropeptide systems involved in social interaction. Although the prolonged effects of early positive soci...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1962157

Peripheral physiological decoding of arousal in a boundary-avoidance task: signal validity and individualized control

Anirudh Natarajan and Paul Sajda

2026-11-03 1 min 284 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Scoring well on a task-demand label does not show that a decoder measures arousal. We ask whether peripheral physiology decodes demand better than EEG, whether the index behaves like graded arousal rather than a clock, and what a controller could do with it. Approach. We re-analysed a pub...

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GROND: automated characterization of periodic and rhythmic EEG patterns

Jin Jing, Chenxi Sun, Tianyu Zhang, Matthew Byrd, Alexandra–Maria Tăuţan, Lara Basovic, Peter N Hadar, Marta P Fernandes, Daniel Goldenholz, Jennifer Kim, Aaron F Struck, Sahar F Zafar and M Brandon Westover

2026-11-03 2 min 469 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Periodic discharges (PDs) and rhythmic delta activity (RDA) are common electroencephalographic patterns in critically ill patients that require detailed characterization—including lateralization, spatial localization, and frequency estimation—according to the American Clinical Neurophysio...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb331

Transformer networks are robust learners of source localization for EEG measurements

David Hart, Simon Polishchuk and Moritz Dannhauer

2026-11-06 1 min 356 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade un...

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ACF-Net: design of an adaptive cross-modal fusion network for cross-subject wearable EEG-EMG gesture recognition

TongLiang Yang

2026-11-06 1 min 292 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Cross-subject gesture recognition in wearable brain–computer interfaces is constrained by four structural deficiencies: symmetric cross-modal attention discarding cortical precedence over muscular execution, priors confined to post hoc analysis, invariance anchored on unimodal features, a...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb723

The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Alon Gorenshtein, Yosef Adiniaev, Tom Liba, Yiftach Barash, Eyal Klang and Oved Daniel

2026-11-06 1 min 297 words JOURNAL NEURAL ENGINEERING

Summary: Objective. A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts. Approach. Retrospective secondary analysis of BigP3BCI 1...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

Junguo Duan
2026-11-07 1 min 72 words LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

👤 Jessie Chin 📅 2026-11-08 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

Jana Zweerings 2026-11-04 1 min 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

👤 Ling Guan 📅 2026-11-05 📖 1 min 📝 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti 📅 2026-11-05 📖 1 min 📝 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

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http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

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http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

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http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

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http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

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http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

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The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

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http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

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http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

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http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

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http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

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Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

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Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

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A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

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Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

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Monthly Updates [November]

📅 2026-11-01 📖 5 min 📝 1013 words FMHY

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RIP, Harvest

👤 zdw 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

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The Qlobe: Rotating Quine (2010)

👤 niceguitars 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

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A Critique of GPUI

👤 ByteAtATime 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.zortax.de/posts/a-critique-of-gpui Comments URL: https://news.ycombinator.com/item?id=50420857 Points: 99 # Comments: 60

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Show HN: Flide: Make Jev-like decision models up to 9.5x faster

👤 unikcc 📅 2026-11-09 📖 1 min 📝 56 words HACKER NEWS

> **Summary:** Decision models need their own inference engine. So we developed Flide, a high-performance and exact inference engine for Jev-like decision models. Flide achieves up to a 9.5x speedup over vLLM and SGLang. It now supports 30+ popular local Jev-like decision models. Start to make your deci...

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https://github.com/OmniJev/Flide

WindyMaps – Free offline maps, 1h/1km weather and route trip planner

👤 alesmikes 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://windymapy.com Comments URL: https://news.ycombinator.com/item?id=50421899 Points: 7 # Comments: 0

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Is Opus 5.5 nerfed? Thinking patterns changed on November 3

👤 rzk 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

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Next-Concept Modeling: Fast, Parallel Coarse-to-Fine Text Generation

👤 theanonymousone 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

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PixelOS

👤 emilburzo 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

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ChatGPT Desktop forwards your SSH agent to remote machines

👤 DenGorbachev 📅 2026-11-09 📖 1 min 📝 38 words HACKER NEWS

> **Summary:** The agent running on a remote machine can authenticate using your local SSH credentials. That means it can push to your private repos on GitHub, for example. Proof: https://github.com/openai/codex/pull/28131#discussion_r3410013180 <hr /> Comments URL: <a href="https://news.ycombinator....

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

> **Summary:** Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

👤 Kehinde Ross 📅 2026-11-06 📖 1 min 📝 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

👤 Junguo Duan
📅 2026-11-07 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton
2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong
2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

Jianqi Li 2026-11-05 1 min 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

Sang Wook Hong 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

Theodore A Kung
2026-11-05 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Monthly Updates [October]

2026-10-01 5 min 1097 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

🔗 **Read full article:**
https://fmhy.net/posts/oct-2026

Monday Daily Thread: Project ideas!

👤 /u/ AutoModerator 📅 2026-11-09 📖 1 min 📝 244 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Project Ideas 💡</h1> Welcome to our weekly Project Ideas thread! Whether you're a newbie looking for a first project or an expert seeking a new challenge, this is the place for you. <h2>How it Works:</h2> Suggest a Project</st...

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https://www.reddit.com/r/Python/comments/1xrd05k/monday_daily_thread_project_ideas/

KetchUp – Open-source SketchUp alternative built with Rust and WebGPU

👤 dcoulmbe 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ketchup3d.github.io/app/ Comments URL: https://news.ycombinator.com/item?id=50418176 Points: 211 # Comments: 57

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Chimera Vocal Synth

👤 nmstoker 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://chimera-vocal-synth.vercel.app/ Comments URL: https://news.ycombinator.com/item?id=50418182 Points: 89 # Comments: 14

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Herb Sutter: When a language's creator and its owners strongly disagree [pdf]

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I'm Not Convinced the AI Bubble Exists

👤 matteoraso 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

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Unlocking any car from X brand in most of the world

👤 infosecau 📅 2026-11-09 📖 1 min 📝 13 words HACKER NEWS

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Learning dynamic affinity diffusion for multi-contrast MRI super-resolution via implicit neural representations

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Huajie Wen, Lixin Liang, Rashid Khan, Inayatul Haq, Kai Chen, Lin Lin, YongSheng Liang, Bingding Huang

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https://www.sciencedirect.com/science/article/pii/S1053811926006397?dgcid=rss_sd_all

Forniceal deep brain stimulation improves memory in Rett syndrome mice by promoting the maturation of newborn hippocampal neurons

Qi WangBin TangZhenyu WuHuda Y. ZoghbiJianrong Tangahttps://ror.org/05cz92x43Jan and Dan Duncan Neurological Research Institute at Texas Children's Hospital, Houston, TX 77030bhttps://ror.org/02pttbw34Department of Pediatrics, Division of Neurology, Baylor College of Medicine, Houston, TX 77030chttps://ror.org/02pttbw34Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030dhttps://ror.org/02pttbw34Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030ehttps://ror.org/02pttbw34Department of Neuroscience, Baylor College of Medicine, Houston, TX 77030fhttps://ror.org/02pttbw34HHMI, Baylor College of Medicine, Houston, TX 77030

📅 2026-10-28 📖 1 min 📝 52 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDeep brain stimulation (DBS) has long been known for its efficacy in mitigating movement disorders. More recently, it has shown promise for addressing disorders of learning and memory: for example...

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https://www.pnas.org/doi/abs/10.1073/pnas.2535887123?af=R

Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2526984123?af=R

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

👤 Jisu Elsa Jacob
📅 2026-10-13 📖 1 min 📝 229 words
FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Editorial: Advances in perceptual decision making and brain oscillations

👤 Francesco Di Gregorio
📅 2026-11-03 📖 1 min 📝 0 words
FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2023605

Exploratory evaluation of electroencephalogram-derived features for classification of mild cognitive impairment in two independent datasets

👤 Makoto Nakane 📅 2026-11-03 📖 1 min 📝 347 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A proper diagnosis of mild cognitive impairment (MCI) is one of the important challenges in the early detection of dementia. For the classification of MCI patients and healthy controls, some electroencephalogram (EEG)-derived indices were calculated and compared. Some studies have been conducted on ...

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Editorial: Machine learning integration in computational neuroscience: enhancing neural data decoding and prediction

👤 George Giakos 📅 2026-11-03 📖 1 min 📝 0 words

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2030505

Short excitatory projections promote burst activity and modular functional organization in neural networks

Claus C. Hilgetag
2026-11-03 1 min 165 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Physical constraints in the brain favor short, distance-dependent connections. However, short-range projections are not merely a structural compromise; they also serve essential functional roles. Here, we explore networks with balanced excitation and inhibition, demonstrating that short excitatory p...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1965686

Persistent memory through triple-loop consolidation under stochastic unit turnover

Jianwei Lou
2026-11-03 1 min 366 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Dissipative cognitive architectures maintain computation through continuous energy expenditure, where units that exhaust their energy are stochastically replaced with fresh random state. This creates a fundamental challenge: how can persistent, context-specific memory survive when all learnable stat...

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Temporal dissociation across the eye-brain-muscle cascade delineates a preparation window before muscle activation in subacute stroke

Jing Tao 2026-11-06 1 min 238 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Upper-limb motor impairment after stroke is compounded by a disrupted sensorimotor cascade linking visual attention, cortical motor preparation, and peripheral muscle activation. Whether these stages are temporally dissociated in subacute stroke, and whether their timing carries information about th...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2002490

Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams 2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1960197

From beat to syntax: evolutionary and developmental paths of a shared neural resource for hierarchical processing in music and language

Tao Gong 2026-11-03 1 min 268 words FRONTIERS NEUROSCIENCE

Summary: Children who track a musical beat well tend to have stronger grammatical skills, and brief exposure to a regular rhythm can transiently improve grammatical processing. These findings raise the possibility of a shared underlying mechanism, and with it two questions: where did the link come from, and ...

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Vagus nerve stimulation in obesity: a systematic review and meta-analysis

Matthew A. Schiefer 2026-11-03 1 min 333 words FRONTIERS NEUROSCIENCE

Summary: BackgroundVagus nerve stimulation (VNS) has been investigated as a potential therapy for obesity because of the vagus nerve’s role in satiety, gastrointestinal signaling, autonomic regulation, and metabolic homeostasis. However, studies vary substantially in stimulation site, interface, parameters, ...

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Editorial: Expanding the therapeutic dose range of cannabinoid-based therapy

Babak Baban 2026-11-03 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2028326

Analysis and improvement of surface electromyography-based American Sign Language hand gesture recognition

Yasuharu Koike 2026-11-03 1 min 243 words FRONTIERS NEUROSCIENCE

Summary: Surface electromyography (sEMG) allows sign language gestures to be recognized from forearm muscle activity without visual access to the hand. Gestures that produce similar forearm activation patterns, however, remain difficult to distinguish. This study examined how these similarities relate to cla...

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REFS: robust EEG feature selection with incomplete labels for multi-dimensional emotion recognition

Xueyuan Xu, Wenjia Dong, Fulin Wei and Li Zhuo

2026-11-06 1 min 268 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Compared to single-type features, multi-type electroencephalography (EEG) features provide a multi-level representation for analyzing multi-dimensional emotions. However, the high dimensionality of multi-type EEG features, combined with the relatively small number of high-quality EEG samp...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb97e

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

👤 Zhixian Yang 📅 2026-11-06 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

👤 Kehinde Ross 📅 2026-11-06 📖 1 min 📝 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

👤 Junguo Duan
📅 2026-11-07 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

👤 Liu Yang 📅 2026-08-19 📖 1 min 📝 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

👤 Hideaki Onishi 📅 2026-09-25 📖 1 min 📝 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

Christoffer Olsson 2026-11-05 1 min 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

Jing-Ning Zhu 2026-11-05 1 min 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

Jianqi Li 2026-11-05 1 min 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

Sang Wook Hong 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

👤 Chunxiao Wan
📅 2026-11-05 📖 1 min 📝 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

Azfar Khalid 2026-11-05 1 min 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

👤 Theodore A Kung
📅 2026-11-05 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

Ximo Wang 2026-11-05 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

Johnsi Inbakumari 2026-11-05 1 min 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

Marie-Anne Rameix-Welti 2026-11-06 1 min 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

Xiao-Hong Li 2026-11-06 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Two Python examples of long-lived unused code: what would justify keeping them?

/u/ BiteCode_Dev 2026-11-08 2 min 441 words REDDIT PYTHON

Summary: <!-- SC_OFF --> I’m working on BiteCode’s Code Roaster series. These are two small source-review examples about You Ain’t Gonna Need It (YAGNI). The evidence is below, so there’s no need to visit a blog to follow the discussion. I’m not treating an old comment as proof of a ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xr6raz/two_python_examples_of_longlived_unused_code_what/

Giving back to the community - The Complete Backend Development Course

/u/ OrganizationGreat823 | 2026-11-08 | 1 min | 79 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Hey everyone, I decided to make my course free in order to help people.
 This course is my backend development course which is about SQL, Python, APIs, Docker, Kubernetes, Linux, Git &...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xr8mnu/giving_back_to_the_community_the_complete_backend/

Playing Zork 1 on a CP/M emulator on Linux (2021)

linuxkernal | 2026-11-08 | 1 min | 13 words | HACKER NEWS

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Playing with the STM32MP2

👤 dverlaeckt80 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

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Dux: Free, open-source Wayland desktop shell with file manager and theming

👤 indigodaddy 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

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Outlook.com Is Quietly Rejecting Email from Domains Without DMARC

👤 xmddmx 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

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The Web Is More Than Enough, by Paul Ford

👤 edward 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ftrain.com/aboard-the-web-is-more-than-enough Comments URL: https://news.ycombinator.com/item?id=50416456 Points: 27 # Comments...

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Starship Flight 15 on 11/13 plans to catch both booster and ship

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Nvidia Board of Directors Is Investigating Jensen Huang

👤 vincentchau 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nvidianews.nvidia.com/news/statement-from-nvidia-board-of-directors Comments URL: <a href="https://news.ycombinator.com/item?id=50416801">https://news.ycombinator.com/item?id=5041...

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Ask HN: What are you working on? (November 2026)

👤 david927 📅 2026-11-08 📖 1 min 📝 23 words HACKER NEWS

> **Summary:** What are you working on? What have you been curious about lately? <hr /> Comments URL: https://news.ycombinator.com/item?id=50417407 Points: 48 # Comments: 225

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Nvidia CEO Jensen Huang's letter to staff about affair

👤 karimf 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.axios.com/2026/11/08/nvidia-ceo-jensen-huang-affair-email-to-company Comments URL: <a href="https://news.ycombinator.com/item?id=50418016">https://news.ycombinator.com/it...

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Easily download files from the Open Science Framework with Papercheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-22 📖 3 min 📝 765 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Aquaporin channels in cytotoxic brain edema following severe head trauma: Systematic review of molecular mechanisms and therapy

1 min 20 words

NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Clayton Rawson, Moneebah Ashraf, Srivardhan Chirasani, Dorsa Farahnak, Michael Karsy, Mehrdad Pehlevani

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Behavioral and neurobiological effects of vicarious social defeat stress in rodents: a scoping review

📖 1 min 📝 27 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Welington dos Santos Silva, Amanda Peixoto Silva, Luciano Lazaro Vieira Lima, Luiz Antonio Chaves Viana, Maria Goretti Silva Brito

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Bilirubin in depression: redox-inflammatory mechanisms, clinical associations, and biomarker potential

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Aarti Ashok Khutale, Prerana Deepak Dalvi, Pratik Prashant Doshi, Aamod Mahesh Pathak, Swathi Suresh

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Establishment of an animal model of GPIAS-defined tinnitus-like behavior with normal auditory brainstem response threshold profiles

📖 1 min 📝 24 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Rong Wang, Yangyang Yu, Qianru Yu, Rui Guo, Menghua Li, Yang Li, Shusheng Gong, Ke Liu

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Corrigendum to “Neuroprotective effects of lixisenatide and liraglutide in the MPTP mouse model of Parkinson’s disease” [Neuroscience 303 (2015) 42–50]

📖 1 min 📝 20 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): W. Liu, J. Jalewa, M. Sharma, G. Li, L. Li, C. Hölscher

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA⁺ and Glx

📖 1 min 📝 34 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Siyuan Fang, Yihui Cheng, Ying Tang, Yupeng Wu, Yulin Wen, Qian Wu, Meiyan Lin, Jianing Kang, Huifang He, Huanping Wang, Gaiying Li, Ying Shen, Jianqi Li

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Temporal coupling shapes the effects of prolonged muscle-tendon vibration on sensorimotor EEG activity during motor imagery

📖 1 min 📝 21 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Nicolas Amiez, Aymeric Guillot, Florian Monjo, Florian Marchand, Sébastien Daligault, Franck Di Rienzo

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https://www.sciencedirect.com/science/article/pii/S1053811926006452?dgcid=rss_sd_all

Same networks, different dynamics: parsimony in auditory cognitive task processing

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Divesh Thaploo, Adebiyi Sobitan, Kristen Wingert, Atsuko Kurosu, Nadia Biassou

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https://www.sciencedirect.com/science/article/pii/S1053811926006233?dgcid=rss_sd_all

Safety of lecanemab in real-world patients with Alzheimer's disease: a prospective, observational case series supported by FAERS signal analysis

👤 Erhao Bao 📅 2026-10-02 📖 1 min 📝 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer's disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

Kay A. Robbins 2026-10-12 1 min 0 words FRONTIERS NEUROINFORMATICS

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https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

Virtual brain models: a roadmap towards effective brain digital twins

👤 Viktor Jirsa 📅 2026-10-29 📖 1 min 📝 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

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Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer's disease analysis

👤 Arshad Ali 📅 2026-10-29 📖 1 min 📝 287 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer's disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

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Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

Lijuan Chen 2026-10-29 1 min 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

👤 Kehinde Ross 📅 2026-11-06 📖 1 min 📝 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

👤 Junguo Duan
📅 2026-11-07 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew
2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada
2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

Christoffer Olsson 2026-11-05 1 min 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

Jing-Ning Zhu 2026-11-05 1 min 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

👤 Ling Guan 📅 2026-11-05 📖 1 min 📝 6 words TDCS TACS TRNS

> **Summary:** [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti 📅 2026-11-05 📖 1 min 📝 41 words TDCS TACS TRNS

> **Summary:** CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

👤 Theodore A Kung
📅 2026-11-05 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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PyPI's redesigned project pages with clearer security signals are now live

/u/AlSweigart 2026-11-08 1 min 81 words REDDIT PYTHON

Summary: <!-- SC_OFF --> https://blog.pypi.org/posts/2026-11-06-ui-updates-complete/ In July, we announced a series of planned updates to the PyPI user interface. Shaped by user research with members of the Pyt...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xr49cg/pypis_redesigned_project_pages_with_clearer/

How to prove equations using diagrams, part 1

👤 frostbyte7 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.epatters.org/post/e-graphs-1/ Comments URL: https://news.ycombinator.com/item?id=50412830 Points: 30 # Comments: 3

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https://www.epatters.org/post/e-graphs-1/

I'm not a fan of Deno's security model (2024)

👤 maxloh 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://goldstein.lol/posts/deno-security/ Comments URL: https://news.ycombinator.com/item?id=50413769 Points: 40 # Comments: 19

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https://goldstein.lol/posts/deno-security/

UK House of Lords to propose mandatory CSAM detection software on smartphones

👤 iamnothere 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://observer.co.uk/news/national/article/fight-to-block-child-sexual-abuse-images-unites-peers-and-mps-in-cross-party-campaign C...

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A field guide to CAPTCHA types and how they break

👤 vasylsmutok 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/vsmutok/FuckCaptcha Comments URL: https://news.ycombinator.com/item?id=50415078 Points: 19 # Comments: 5

🔗 **Read full article:**
https://github.com/vsmutok/FuckCaptcha

A Little Snitch for Windows

👤 BadBoyBubby 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://packethound.app/ Comments URL: https://news.ycombinator.com/item?id=50415135 Points: 21 # Comments: 6

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https://packethound.app/

Ads in Popular Apps Give Police a Powerful Tool to Spy on Users

👤 mgh2 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.bloomberg.com/features/2026-police-phone-tracking-advertising-tech/ Comments URL: <a href="https://news.ycombinator.com/item?id=50415281">https://news.ycombinator.com/item...

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https://www.bloomberg.com/features/2026-police-phone-tracking-advertising-tech/

Goldman Sachs Is Pulling Claude Code from Its Bankers and Traders

👤 TheWiggles 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.thestateofai.com/news/goldman-sachs-claude-code-traders-bankers Comments URL: <a href="https://news.ycombinator.com/item?id=50415628">https://news.ycombinator.com/item?id=5041...

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https://www.thestateofai.com/news/goldman-sachs-claude-code-traders-bankers

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

Adriel Carridice 2025-11-04 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Disrupted imagery, disrupted scenes: mental scene construction in behavioural-variant frontotemporal dementia

1 min 18 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Mohamad El Haj, Guillaume Chapelet, Dimitrios Kapogiannis, Claire Boutoleau-Bretonnière

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600483X?dgcid=rss_sd_all

Paclitaxel induces region-specific microglial remodeling associated with chronic behavioral deficits and modulated by electroacupuncture in rats

📖 1 min 📝 19 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Shinya Hiraiwa, Taiga Muramoto, Masahiro Umeda, Satomi Ebara, Fumihiko Fukuda

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004828?dgcid=rss_sd_all

Gastrin-releasing peptide in the anterior medial nucleus accumbens or paraventricular nucleus suppresses chow intake under food restriction but not palatable diet intake in nonrestricted rats

📖 1 min 📝 28 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Regan Cochran, Ella Irvin, Yiwei Tang, Stacey Chuang, Adele Nadeau, Smruti Patel, Lindsey Wilson, Weijia Zhang, Wayne E. Pratt

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004877?dgcid=rss_sd_all

SREBP1 inhibition downregulates NR2B-PSD95 interaction in glutamate excitotoxicity following cerebral ischemic reperfusion injury

📖 1 min 📝 16 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Lopmudra P. Sarode, Bhavesh Verma, Rajesh Ugale

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004841?dgcid=rss_sd_all

Individualized structural brain network and transcriptomic characterization of heterogeneity in type 2 diabetes mellitus

📖 1 min 📝 36 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Haolin Wang, Jinglei Xu, Yue Wu, Xin Du, Yang Zhang, Ying Zhai, Zhihui Zhang, Qian Wu, Minghuan Lei, Qi An, Wenjie Cai, Yanmin Peng, Feng Liu, Quan Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007207?dgcid=rss_sd_all

Correcting automatic decisions

Luis A. Mejia 2026-11-02 1 min 11 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02510-x Correcting automatic decisions

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02510-x

Misfolded tau propagates via templated seeding

Qingzhong Ren 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02512-9 Misfolded tau propagates via templated seeding

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02512-9

Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-11-02 1 min 22 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02530-7 Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02530-7

Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

Roman J. Giger 2026-11-05 1 min 21 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 05 November 2026; doi:10.1038/s41593-026-02534-3 Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02534-3

Ddx3x deficiency in mice impairs corticofugal motor circuits and causes motor deficits that are attenuated by early-life exercise

👤 Lauren DierdorffIkrame El KhaldiMarta Garcia-FornWei WangJeronimo LukinCorinne M. SmithAdele MossaZhuhao WuSilvia De Rubeisahttps://ror.org/04a9tmd77Seaver Autism Center for Research and Treatment, Icahn School of Medicine at Mount Sinai, New York, NY 10029bhttps://ror.org/04a9tmd77Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY 10029chttps://ror.org/04a9tmd77Department of Neuroscience, Icahn School of Medicine at Mount Sinai, New York, NY 10029dhttps://ror.org/04a9tmd77The Mindich Child Health and Development Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029ehttps://ror.org/04a9tmd77The Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029fhttps://ror.org/04a9tmd77The Alper Center for Neural Development and Regeneration Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029ghttps://ror.org/02r109517Helen & Robert Appel Alzheimer's Disease Research Institute, Feil Family Brain and Mind Research Institute, Weill Cornell Medicine, New York, NY 10021hhttps://ror.org/04a9tmd77Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, NY 10029ihttps://ror.org/04a9tmd77Department of Graduate Education, Icahn School of Medicine at Mount Sinai, New York, NY 10029

📅 2026-10-29 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAcquisition of motor skills during infancy is critical for a child's ability to interact with the physical and social environment and motor impairments are frequent in individuals with neurodevelo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535732123?af=R

Respiratory phase orchestrates human olfactory cortical dynamics

👤 Alisa Y. HathawayTodd P. Colemanahttps://ror.org/00f54p054Department of Electrical Engineering, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Department of Bioengineering, Stanford University, Stanford, CA 94305

📅 2026-11-06 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceDiagnosing smell loss in patients with cognitive disorders remains challenging because conventional tests demand complex responses that measure memory and literacy instead of basic sensory functio...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619920123?af=R

The purinergic receptor P2Y13 controls the activation and mode of division of adult ventricular–subventricular zone neural stem cells

Lucía Paniagua-HerranzJulia Serrano-LópezCelia Llorente-SáezMarina Leonor Pérez-SanzDavid de Agustín-DuránPere Duart-AbadiaAna Domingo-MuelasÁfrica Vincelle-NietoPaloma BragadoDiana Manzano-FrancoMarina Arribas-BlázquezLucía GallegoAraceli Garcia-MoraLuis Alcides Olivos-OréSergio GascónÁlvaro Gutierrez-UzquizaBenedikt BerningerRaquel Pérez-SenEsmerilda G. DelicadoRosa Gómez-VillafuertesArmando Reyes-PalomaresAntonio R. ArtalejoIsabel FariñasFelipe Ortegaahttps://ror.org/02p0gd045Departamento de Bioquímica y Biología Molecular, Universidad Complutense de Madrid, Madrid 28040, SpainbInstituto Universitario de Investigación en Neuroquímica, Departamento de Bioquímica y Biología Molecular, Madrid 28040, Spainchttps://ror.org/014v12a39Instituto de Investigación Sanitaria del Hospital Clínico San Carlos de Madrid, Madrid 28040, Spaindhttps://ror.org/02p0gd045Departamento de Farmacología y Toxicología, Facultad de Veterinaria, Universidad Complutense de Madrid, Madrid 28040, Spainehttps://ror.org/043nxc105Departamento de Biología Celular y Biología Funcional, Universidad de Valencia, Valencia 46100, Spainfhttps://ror.org/043nxc105Instituto de Biotecnología y Biomedicina, Departamento de Genética, Facultad de Biología, Universidad de Valencia, Valencia 46100, Spainghttps://ror.org/02g87qh62Centro de Investigación Biomédica en Red sobre Enfermedades Neurodegenerativas, Departamento de Biología Celular y Biología Funcional, Valencia 46100, Spainhhttps://ror.org/02gfc7t72Cajal Neuroscience Center, Laboratory of direct reprogramming, Spanish National Research Council, Madrid 28805, Spainihttps://ror.org/0220mzb33Centre for Developmental Neurobiology, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomjhttps://ror.org/0220mzb33Medical Research Council Centre for Neurodevelopmental Disorders, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomkhttps://ror.org/023b0x485Institute of Physiological Chemistry, University Medical Center Johannes Gutenberg University, Mainz 55128, GermanylFocus Program Translational Neuroscience, Johannes Gutenberg University, 55128, Mainz, Germanymhttps://ror.org/036b2ww28Department of Molecular Biology and Biochemistry, Faculty of Science, University of Malaga, Malaga SE1 1UL, Spain

📅 2026-11-04 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceAdult neural stem cells (NSCs) sustain lifelong neurogenesis in the mammalian brain, yet the mechanisms controlling their activation and self-renewal remain poorly understood. We identify the puri...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2618940123?af=R

Dissociation of gravity and limb dynamics in motor control

👤 Valeriya GritsenkoTjaša KunavarMarko JamšekSergiy YakovenkoOlivier WhiteJan Babičahttps://ror.org/011vxgd24Departments of Human Performance/Neuroscience, West Virginia University, Rockefeller Neuroscience Institute, Morgantown, WV 26506bLaboratory for Neuromechanics and Biorobotics, Department of Automatics, Biocybernetics, and Robotics, Jožef Stefan Institute, Ljubljana 1000, Sloveniachttps://ror.org/02vjkv261Université Bourgogne Europe, Institut national de la santé et de la recherche médicale, Cognition, Action and Sensorimotor Plasticity, Unité Mixte de Recherche 1093, Dijon 21000, France

📅 2026-11-03 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceThe nervous system must control movement in presence of variable intrinsic and extrinsic forces, yet it has remained unclear whether mechanically equivalent forces arising from gravity and limb dy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2613283123?af=R

Lead impairs long-term memory by disrupting mitochondrial redox homeostasis in Drosophila

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📅 2026-11-04 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceEnvironmental lead exposure is strongly associated with cognitive deficits, yet how it disrupts memory at the cellular level remains unclear. Here, we show that lead selectively impairs protein-sy...

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Hypothalamic lncRNA Synage prevents age-related sperm quality decline by suppressing A-type potassium currents in GnRH neurons

Fei WangMengyi ZhangMengtian WangShuhan TianQian BianYaohui HeShuiqiao YuanQinghua ShiWen LiuJun XiaoLi WangJi LiuXiaoyuan Songahttps://ror.org/04c4dkn09Department of Urology, The First Affiliated Hospital of University of Science & Technology of China, Hefei National Research Center for Physical Sciences at the Microscale, Ministry of Education Key Laboratory for Cellular Dynamics, Center for Advanced Interdisciplinary Science and Biomedicine of the Institute of Health and Medicine, Division of Life Sciences and Medicine, University of Science and Technology of China, Hefei, Anhui 230026, Chinabhttps://ror.org/04c4dkn09Division of Life Sciences and Medicine, School of Biomedical Engineering, University of Science and Technology of China, Hefei, Anhui 230026, Chinachttps://ror.org/04c4dkn09School of Biomedical Engineering, Suzhou Institute for Advanced Research, University of Science and Technology of China, Suzhou, Jiangsu 215123, Chinadhttps://ror.org/03hz5th67Department of Neuroscience, Faculty of Life and Health Sciences, Shenzhen University of Advanced Technology, Shenzhen, Guangdong 518107, Chinaehttps://ror.org/00mcjh785State Key Laboratory of Cellular Stress Biology, School of Pharmaceutical Sciences, Xiamen University, Xiamen, Fujian 361102, Chinafhttps://ror.org/00p991c53Institute of Reproductive Health, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei 430030, Chinaghttps://ror.org/04c4dkn09Department of Urology, Anhui Provincial Hospital, The First Affiliated Hospital of University of Science & Technology of China, Division of Life Sciences and Medicine, University of Science & Technology of China, Hefei, Anhui 230001, Chinahhttps://ror.org/0220qvk04Department of Assisted Reproduction, Shanghai Ninth People's Hospital, Shanghai JiaoTong University School of Medicine, Shanghai 200011, ChinaiInstitute of Artificial Intelligence, Hefei Comprehensive National Science Center, Hefei 230026, China

📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceMen increasingly delay fatherhood, yet paternal aging lowers sperm quality and increases health risks to offspring. The mechanisms underlying these reproductive changes remain incompletely defined...

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https://www.pnas.org/doi/abs/10.1073/pnas.2612408123?af=R

Transcranial magnetic stimulation to the dorsolateral prefrontal cortex modulates single-neuron activity in humans

👤 Charles W. DickeyUmair HassanHiroto KawasakiAriane E. RhoneChristopher C. ClineMatthew A. HowardNicholas T. TrappAaron D. BoesJoel I. BergerCorey J. Kellerahttps://ror.org/00f54p054Department of Psychiatry and Behavioral Sciences, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Wu Tsai Neurosciences Institute, Stanford University, Stanford, CA 94305chttps://ror.org/036jqmy94Department of Neurosurgery, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Department of Psychiatry, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Department of Neurology, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Department of Pediatrics, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Iowa Neuroscience Institute, Carver College of Medicine, University of Iowa, Iowa City, IA 52242hVeterans Affairs Palo Alto Healthcare System, and the Sierra Pacific Mental Illness Research, Education, and Clinic Center, Palo Alto, CA 94305

📅 2026-11-02 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceTranscranial magnetic stimulation (TMS) to the dorsolateral prefrontal cortex is a common treatment for depression. However, it remains unknown how TMS affects single neurons in deep brain circuit...

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https://www.pnas.org/doi/abs/10.1073/pnas.2610803123?af=R

Hepatocyte-derived exosomal miR-122 mediates liver-to-brain crosstalk and neuroinflammation in alcohol-exposed Alzheimer’s disease mice

👤 Veronika BrezaniMarti Ortega-RiberaNeda RezagholizadehRadhika JoshiViliam BrezaniMrigya BabutaBruna de Araujo Cardoso dos SantosAdam ZivnyPrashanth Thevkar NageshIonita GhiranElizabeth Michelle ShreyerScott Brian MinchenbergAllyson FlavinGyongyi Szaboahttps://ror.org/04drvxt59Division of Gastroenterology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215bHarvard Medical School, Boston, MA 02115chttps://ror.org/04drvxt59Department of Anesthesia, Critical Care and Pain Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215dhttps://ror.org/05a0ya142Broad Institute of MIT and Harvard, Cambridge, MA 02142

📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceExcessive alcohol use damages both the liver and the brain, but how these organs communicate during injury remains unclear. Our study identifies a previously unrecognized liver-to-brain communicat...

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From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne 2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

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Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka 2026-10-23 1 min 318 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

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Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

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Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

AABC Study Group 2026-11-06 1 min 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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A full-hardware edge detection system based on a highly robust dual random number generator

Xiaobing Yan 2026-11-06 1 min 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

Junguo Duan
2026-11-07 1 min 72 words LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

Azfar Khalid 2026-11-05 1 min 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

👤 Theodore A Kung
📅 2026-11-05 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier 2026-11-07 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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A research on the typical images of animals in the dreams of Naxi.

2026-09-10 1 min 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

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https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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https://fmhy.net/posts/sept-2026

WaveDudeNext 2D Wave Simulator

👤 sweetdingoes 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/graealex/WaveDudeNext Comments URL: https://news.ycombinator.com/item?id=50411083 Points: 32 # Comments: 8

🔗 **Read full article:**
https://github.com/graealex/WaveDudeNext

Agent memory needs its OpenTelemetry moment

👤 mthines 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.lorekit.io/blog/open-agent-memory-format Comments URL: https://news.ycombinator.com/item?id=50411467 Points: 81 # Comments: ...

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https://www.lorekit.io/blog/open-agent-memory-format

Llama-modchip: 284B DeepSeek on 64GB RAM, no GPU

👤 joshcsimmons 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/synchopate/llama-modchip Comments URL: https://news.ycombinator.com/item?id=50411961 Points: 59 # Comments: 24

🔗 **Read full article:**
https://github.com/synchopate/llama-modchip

JIT bitmap text rendering for microcontrollers

👤 dmit 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://linebender.org/blog/jit-bitmap-text/ Comments URL: https://news.ycombinator.com/item?id=50412072 Points: 35 # Comments: 6

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https://linebender.org/blog/jit-bitmap-text/

Petered (what happens after the Peter Principle)

👤 adrianhoward 📅 2026-11-08 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://aworkinglibrary.com/writing/petered Comments URL: https://news.ycombinator.com/item?id=50412152 Points: 79 # Comments: 37

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Person of Interest

krthr 2026-11-08 1 min 13 words HACKER NEWS

> **Summary:** Article URL: https://krthr.co/person-of-interest/ Comments URL: https://news.ycombinator.com/item?id=50412318 Points: 140 # Comments: 82

🔗 **Read full article:**
https://krthr.co/person-of-interest/

There are 13 ways to analyse a replication study, but only one of them is coherent.

noreply@blogger.com (Daniel Lakens) 2026-05-03 6 min 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice 2026-09-23 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

🔗 **Read full article:**
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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

🔗 **Read full article:**
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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/41703344/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108015845&v=2.20.1

EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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https://pubmed.ncbi.nlm.nih.gov/42538927/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261108015845&v=2.20.1

Cognition in healthy aging, ages 36 to 108: Associations with dementia risk factors and neurodegenerative disease biomarkers

👤 AABC Study Group
📅 2026-11-06 📖 1 min 📝 74 words LOW VISION

Summary: Most evidence on dementia risk factors comes from older or clinical samples, leaving their relation to normal cognitive aging unclear. In the Aging Adult Brain Connectome cohort (N = 1,396; ages 36-108; 2,878 visits; mean follow-up 5.0 years), we tested eighteen risk factors against three cognitive ...

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https://pubmed.ncbi.nlm.nih.gov/42990508/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261108015842&v=2.20.1

A full-hardware edge detection system based on a highly robust dual random number generator

👤 Xiaobing Yan
📅 2026-11-06 📖 1 min 📝 65 words LOW VISION

Summary: Conventional edge computing hardware at autonomous driving terminals is limited by chip area and power efficiency for complex image data. Recently, researchers have used memristors to construct dedicated hardware systems, achieving advancements in multiple image processing fields such as pattern rec...

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Large Vision-Language Models for Pediatric Motor Tic Detection: Case-Control Diagnostic Study

Zhixian Yang 2026-11-06 1 min 52 words LOW VISION

Summary: CONCLUSIONS: These findings suggest that a vision-language model framework can capture clinically relevant tic features beyond conventional geometric descriptors. By combining high sensitivity (recall) with text-based diagnostic justifications aligned with clinical severity dimensions (eg, frequency...

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Uplifting RNA Science in Africa: A Call to Action

Kehinde Ross 2026-11-06 1 min 71 words LOW VISION

Summary: The 21st century has witnessed the emergence of an entire new class of therapeutic and prophylactic medicines based on RNA. These transformative advances have arisen from decades of fundamental and applied research efforts to understand, exploit, and modify the properties of RNA to improve lives and...

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CrossBoneNet: An Ultralightweight Hybrid CNN-ViT Architecture via Asymmetric Reversed Cross-Attention for Radiographic Fracture Detection

👤 Orhan Er 📅 2026-11-06 📖 1 min 📝 60 words LOW VISION

Summary: Misdiagnoses in emergency departments are often associated with the difficulty in interpreting radiological findings on bone fractures due to millimetric fractures, overlapping bones, or superimpositions. Despite the high efficiency of state-of-the-art models in diagnosing such cases, their massive ...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

👤 Junguo Duan
📅 2026-11-07 📖 1 min 📝 72 words

LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-11-05 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: INTRODUCTION: Predictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dM...

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Obsessive-Compulsive Disorder: Comorbidity, Biomarkers, Circuits, and Treatment Strategies

👤 Jing-Ning Zhu 📅 2026-11-05 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Obsessive-compulsive disorder (OCD), characterized by intrusive thoughts and repetitive behaviors, frequently co-occurs with anxiety disorders, contributing to greater diagnostic complexity, more severe symptoms, and poorer treatment outcomes. In this review, we consider OCD in the context of its co...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

Ling Guan 2026-11-05 1 min 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

Rogério Panizzutti 2026-11-05 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

Azfar Khalid 2026-11-05 1 min 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Next-generation control strategies for lower-extremity exoskeletons: State-of-the-art, limitations, and future directions

👤 Theodore A Kung
📅 2026-11-05 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Wearable powered exoskeletons are emerging rehabilitation tools for restoring gait in individuals with neurologic or motor impairments, including stroke, spinal cord injury, myopathy, and cerebral palsy. Lower-extremity exoskeleton-assisted walking can improve motor function, gait parameters, balanc...

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Research advances in brain-computer interfaces in oncology: from cancer pain management to neuroengineering for brain tumors and tumor-neural-immune interactions

👤 Ximo Wang 📅 2026-11-05 📖 1 min 📝 69 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) have made significant progress in fields such as neurorehabilitation, motor reconstruction, and alternative communication by capturing, decoding, and providing feedback on brain activity. In contrast, the application of BCI in oncology has a relatively late start, bu...

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Letter to the Editor: "Bladder Contractility Classification Before Bladder Outlet Surgery: Do BCI, Wmax and the Maastricht-Hannover Nomogram Agree and Does It Matter for Outcome?"

👤 Johnsi Inbakumari 📅 2026-11-05 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

👤 Caili Gong 📅 2026-11-06 📖 1 min 📝 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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